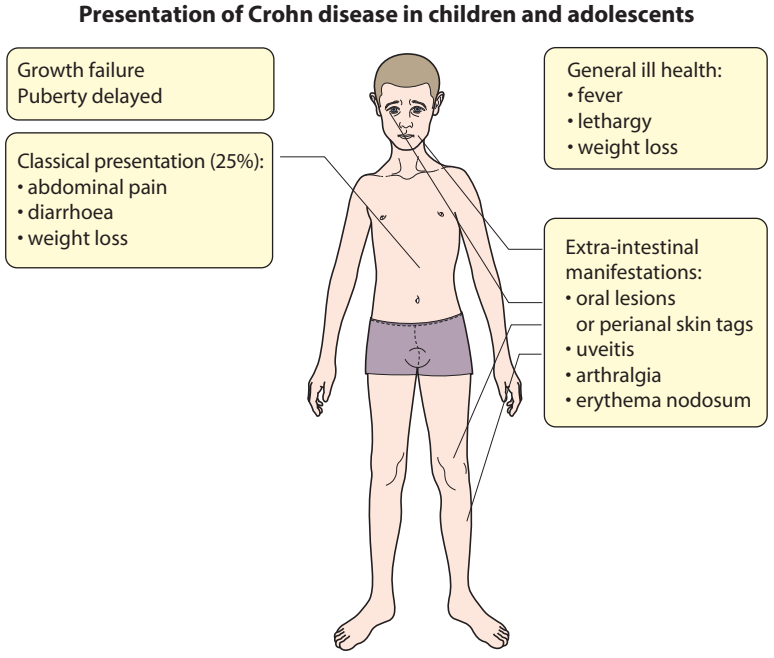


Figure 13.13 Presentation of Crohn disease in children and adolescents.

psychological problems. It may also mimic anorexia nervosa. The presence of raised inflammatory markers (platelet count, ESR and CRP), iron deficiency anaemia and low serum albumin are helpful in both making a diagnosis and confirming a relapse.

Crohn disease is a transmural, focal, subacute or chronic inflammatory disease, most commonly affecting the distal ileum and proximal colon. Initially there are areas of acutely inflamed, thickened bowel. Subsequently, strictures of the bowel and fistulae may develop between adjacent loops of bowel, between bowel and skin or to other organs (e.g. vagina, bladder).

Diagnosis is based on endoscopic and histological findings on biopsy. Upper gastrointestinal endoscopy, ileocolonoscopy and small bowel imaging are required. The histological hallmark is the presence of non-caseating epithelioid cell granulomata, although this is not identified in up to 30% at presentation. Small bowel imaging may reveal narrowing, fissuring, mucosal irregularities and bowel wall thickening.

Remission is induced with nutritional therapy, when the normal diet is replaced by whole protein modular feeds (polymeric diet) for 6–8 weeks. This is effective in 75% of cases. Systemic steroids are required if ineffective.

Relapse is common and immunosuppressant medication (azathioprine, mercaptopurine or methotrexate) may be required to maintain remission. Anti-tumour necrosis factor agents (infliximab or adalimumab) may be needed when conventional treatments have failed. Long-term supplemental enteral nutrition (often with overnight nasogastric or gastrostomy feeds) may be helpful in correcting growth failure. Surgery is necessary for complications of Crohn disease – obstruction, fistulae, abscess formation or severe localised disease unresponsive to medical treatment, often manifesting as growth failure. In general, the long-term prognosis for Crohn disease beginning in childhood is good and most patients lead normal lives, despite occasional relapsing disease.

Growth failure and delayed puberty are features of Crohn disease in children.

Ulcerative colitis

Ulcerative colitis is a recurrent, inflammatory and ulcerating disease involving the mucosa of the colon. Characteristically, the disease presents with rectal bleeding, diarrhoea and colicky pain. Weight loss and growth failure may occur, although this is less frequent than in Crohn disease. Extraintestinal complications include erythema nodosum and arthritis.

The diagnosis is made on endoscopy (upper and ileocolonoscopy) and on the histological features, after exclusion of infective causes of colitis. There is a confluent colitis extending from the rectum proximally for a variable length. In contrast to adults, in whom the colitis is usually confined to the distal colon, 90% of children have a pancolitis. Histology reveals mucosal inflammation, crypt damage (cryptitis, architectural distortion, abscesses and crypt loss) and ulceration. Small bowel imaging is required to check that extracolonic inflammation suggestive of Crohn disease is not present.

In mild disease, aminosalicylates (balsalazide and mesalazine) are used for induction and maintenance therapy. Disease confined to the rectum and sigmoid colon may be managed with topical steroids. More aggressive or extensive disease requires systemic steroids for acute exacerbations and immunomodulatory therapy, e.g. azathioprine to maintain remission alone or in combination with low-dose corticosteroid therapy.

Severe fulminating disease is a medical emergency and requires treatment with intravenous fluids and steroids. If this fails to induce remission, ciclosporin may be used.

Colectomy with an ileostomy or ileorectal pouch is undertaken for severe fulminating disease which may be complicated by a toxic megacolon, or for chronic

poorly controlled disease. There is an increased incidence of adenocarcinoma of the colon in adults (1 in 200 risk for each year of disease between 10 and 20 years from diagnosis). Regular colonoscopic screening is performed after 10 years from diagnosis.

Constipation

Constipation is an extremely common reason for consultation in children. Parents may use the term to describe decreased frequency of defecation; the degree of hardness of the stool and painful defecation. The 'normal' frequency of defecation is highly variable and varies with age. Infants have an average of four stools per day in the first week of life, but this falls to an average of two per day by 1 year of age. Breast-fed infants may not pass stools for several days and be entirely healthy. By 4 years of age, children usually have a stool pattern similar to adults, in whom the normal range varies from three stools per day to three stools per week. A pragmatic definition of constipation is the infrequent passage of dry, hardened faeces often accompanied by straining or pain. There may be abdominal pain which waxes and wanes with passage of stool or overflow soiling (see Chapter 23, Emotions and behaviour).

The cause of constipation is often unclear and multifactorial. In babies, Hirschsprung disease, anorectal abnormalities, hypothyroidism and hypercalcaemia need to be considered. Constipation may be precipitated by dehydration or reduced fluid intake or an anal fissure causing pain. In older children, it may relate to problems with toilet training, unpleasant toilets or stress.

Examination often reveals a palpable abdominal mass in a well-looking child. Digital rectal examination should only be performed by a paediatric specialist and only if a pathological cause is suspected. 'Red Flag' symptoms and signs indicative of more significant pathology are detailed in Box 13.4. Investigations are not usually required to diagnose idiopathic constipation, but are carried out as indicated by history or clinical findings.

Constipation arising acutely in young children, for example after an acute febrile illness, usually resolves spontaneously or with the use of mild laxatives and extra fluids.

In more long-standing constipation, the rectum becomes overdistended, with a subsequent loss of feeling the need to defecate. Involuntary soiling may occur as contractions of the full rectum inhibit the internal sphincter, leading to overflow. Management of these children is likely to be more difficult and protracted (Fig. 13.14). Children of school age are frequently teased as a result and secondary behavioural problems are common.

It should be explained to the child and the parents that the soiling is involuntary and that recovery of normal rectal size and sensation can be achieved but may take a long time. The initial aim is to evacuate the overloaded rectum completely. This can generally be achieved using a disimpaction regimen of stool softeners, initially with a macrogol laxative, e.g. polyethylene glycol + electrolytes (Movicol Paediatric Plain). An escalating dose regimen is administered over 1–2 weeks or until impaction resolves. If this proves unsuccessful, a stimulant laxative, e.g. senna, or sodium picosulphate, may also be required. If the polyethylene glycol + electrolytes is not tolerated, an osmotic laxative can be substituted.

Box 13.4 'Red Flag' symptoms or signs in the child with constipation

'Red Flag' symptom/ signs	Diagnostic concern
Failure to pass meconium within 24 h of life	Hirschsprung disease
Failure to thrive/growth failure	Hypothyroidism, coeliac disease, other causes
Gross abdominal distension	Hirschsprung disease or other gastrointestinal dysmotility
Abnormal lower limb neurology or deformity, e.g. talipes or secondary urinary incontinence	Lumbosacral pathology
Sacral dimple above natal cleft, over the spine – naevus, hairy patch, central pit, or discoloured skin	Spina bifida occulta
Abnormal appearance/ position/patency of anus	Abnormal anorectal anatomy
Perianal bruising or multiple fissures	Sexual abuse
Perianal fistulae, abscesses or fissures	Perianal Crohn disease

Disimpaction must be followed by maintenance treatment to ensure ongoing regular, pain-free defecation. Polyethylene glycol (with or without a stimulant laxative) is generally the treatment of choice. The dose should be gradually reduced over a period of months in response to improvement in stool consistency and frequency.

Dietary interventions alone are unlikely to be successful in managing constipation in this situation, although the child should receive sufficient fluid and a balanced diet including adequate fibre. The child should be encouraged to sit on the toilet after mealtimes to utilise the physiological gastrocolic reflex and improve the likelihood of success.

The outcome is more likely to be successful if the child is engaged in the treatment process. This requires exploring the child's concerns and motivation to change. Sometimes use of a star chart is helpful to record and reward progress, as well as motivating the child.

Encouragement by family and health professionals is essential, as relapse is common and psychological support is sometimes required. Occasionally, the faecal

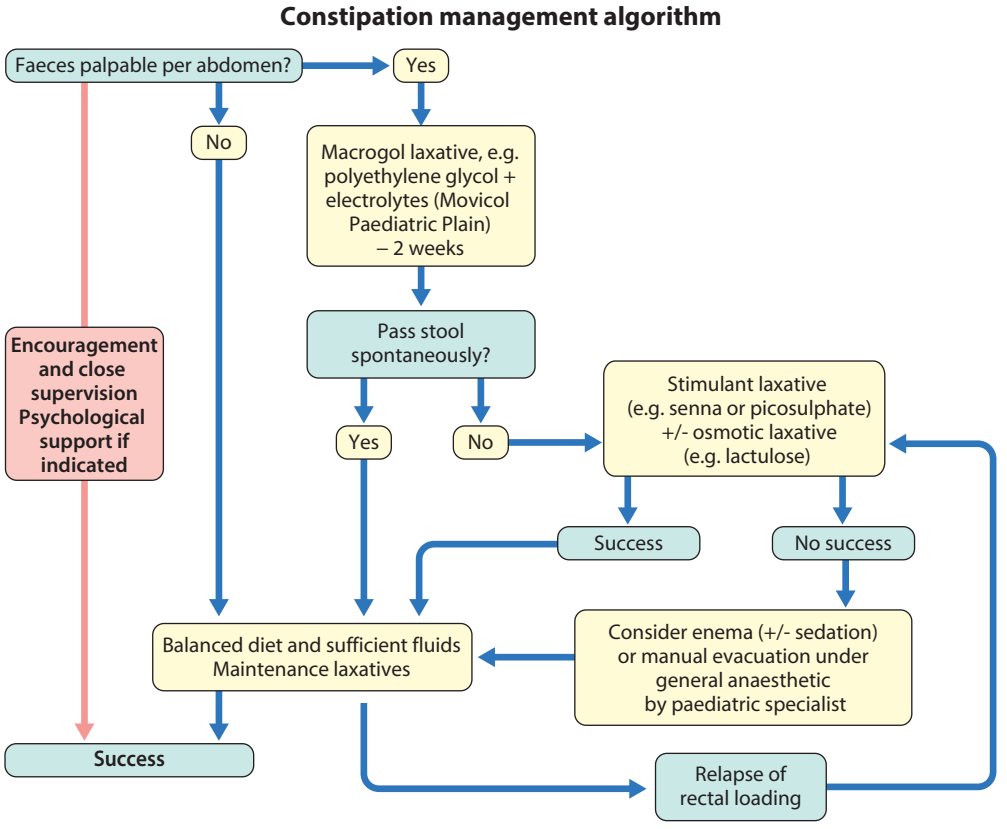


Figure 13.14 Summary of the management of constipation.

retention is so severe that evacuation is only possible using enemas or by manual evacuation under an anaesthetic. They should only be performed under specialist supervision, paying particular attention to avoiding distress and embarrassment for the child.

Hirschsprung disease

The absence of ganglion cells from the myenteric and submucosal plexuses of part of the large bowel results in a narrow, contracted segment. The abnormal bowel extends from the rectum for a variable distance proximally, ending in a normally innervated, dilated colon. In 75% of cases, the lesion is confined to the rectosigmoid, but in 10% the entire colon is involved. Presentation is usually in the neonatal period with intestinal obstruction heralded by failure to pass meconium within the first 24 h of life. Abdominal distension and later bile-stained vomiting develop (Fig. 13.15). Rectal examination may reveal a narrowed segment and withdrawal of the examining finger often releases a gush of liquid stool and flatus. Temporary improvement in the obstruction following the dilatation caused by the rectal examination can lead to a delay in diagnosis.

Occasionally, infants present with severe, life-threatening Hirschsprung enterocolitis during the first few weeks of life, sometimes due to *Clostridium difficile* infection. In later childhood, presentation is with chronic constipation, usually profound, and associated with abdominal distension but usually without soiling. Growth failure may also be present.



Figure 13.15 Abdominal distension from Hirschsprung disease.

Summary

Hirschsprung disease

- Absence of myenteric plexuses of rectum and variable distance of colon
- Presentation – usually intestinal obstruction in the newborn period following delay in passing meconium. In later childhood – profound chronic constipation, abdominal distension and growth failure
- Diagnosis – suction rectal biopsy.

Diagnosis is made by demonstrating the absence of ganglion cells, together with the presence of large, acetylcholinesterase-positive nerve trunks on a suction rectal biopsy. Anorectal manometry or barium studies may be useful in giving the surgeon an idea of the length of the aganglionic segment but are unreliable for diagnostic purposes. Management is surgical and usually involves an initial colostomy followed by anastomosing normally innervated bowel to the anus.

Further reading

NICE: Guideline. Diarrhoea and vomiting in children under 5, 2009.

NICE: Guideline. Coeliac disease, 2009.

NICE: Guideline. Constipation in children and young people, 2010.

Beattie M, Dhawan A, Puntis J: *Paediatric Gastroenterology, Hepatology and Nutrition* (Oxford Specialist Handbooks in Paediatrics), Oxford, 2009, Oxford University Press.

Short handbook

Kleinman RE, Goulet OJ, Mieli-Vergani G, et al: editors: *Walker's Pediatric Gastrointestinal Disease: Physiology, Diagnosis, Management*, ed 5, Ontario, 2008, Decker.

Comprehensive 2-volume textbook.

This page intentionally left blank

14

Infection and immunity

The febrile child	241	Tuberculosis	260
Serious life-threatening infections	243	Tropical infections	262
Specific bacterial infections	248	HIV infection	262
Common viral infections	250	Lyme disease	265
Uncommon viral infections	256	Immunisation	265
Prolonged fever	258	Immune deficiency	268

Infections are the most common cause of acute illness in children.

Worldwide, acute respiratory infections, diarrhoea, neonatal infection, malaria, measles and HIV infection, often accompanied by undernutrition, are responsible for the deaths of more than 4.5 million children <5 years old annually (Fig. 14.1).

In developed countries, morbidity and mortality from infections has declined dramatically, and deaths from infectious diseases are uncommon. However, serious infections still occur, e.g. meningococcal septicaemia, meningitis, and multi-drug resistant pathogens, and some have re-emerged, e.g. tuberculosis and PVL-toxin-secreting *Staphylococcus aureus*, and require early recognition and treatment. Children with immune deficiency are vulnerable to a range of unusual or opportunist pathogens.

With air travel, tropical diseases are encountered in all countries. In addition, epidemics may spread widely, e.g. SARS and H_1N_1, with children (and the elderly) most vulnerable.

The febrile child

Most febrile children have a brief, self-limiting viral infection. Mild localised infections, e.g. otitis media or tonsillitis, may be diagnosed clinically. The clinical problem lies in identifying the relatively few children with a serious infection which needs prompt treatment.

Clinical features

When assessing a febrile child, consider the following.

(i) How is fever identified in children?

Parents usually know if their child has been febrile.

In hospital, it is measured at:

- <4 weeks old by an electronic thermometer in the axilla
- 4 weeks to 5 years by an electronic or chemical dot thermometer in the axilla or infrared tympanic thermometer.

In general, axillary temperatures underestimate body temperature by 0.5°C.

(ii) How old is the child?

Febrile infants <3 months old present with non-specific clinical features (see Box 10.2) and often have a bacterial infection, which cannot be identified reliably on clinical examination alone. It is uncommon for them to have the common viral infections of older infants and children because of passive immunity from their mothers (Fig. 14.2). Unless a clear cause for the fever is identified, they require urgent investigation with a septic screen (Box 14.1) and intravenous antibiotic therapy given immediately to avoid the illness becoming more severe and to prevent rapid spread to other sites of the body. This is considered in more detail in the section on neonatal infection (Chapter 10 Neonatal medicine).

(iii) Are there risk factors for infection?

These include:

- Illness of other family members
- If a specific illness is prevalent in the community
- Unimmunised
- Recent travel abroad, e.g. malaria, typhoid

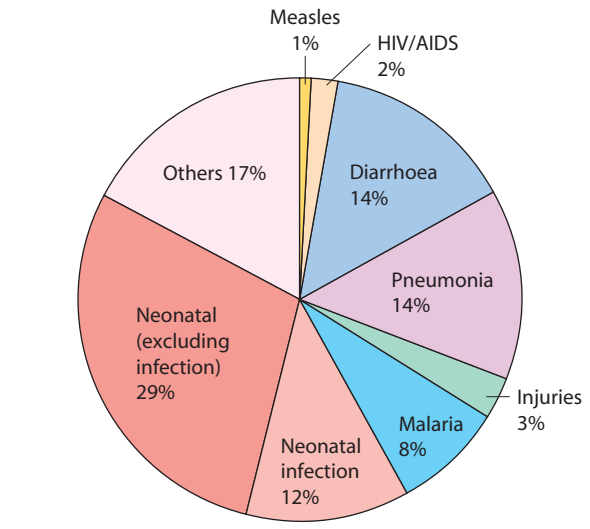


Figure 14.1 Worldwide causes of death in children <5 years old, 2008 (http://www.who.int, Accessed January 2011).

Every year, over half of the 8.8 million deaths of children <5 years old is from infections

Malaria
- Deaths mostly from cerebral malaria from *Plasmodium falciparum* in Sub-Saharan Africa
- Deaths have been reduced in many countries by insecticide-treated bed nets and early treatment with artemisinin-based combination therapy

Diarrhoea
- Most <2 years old
- Often bacterial, although rotavirus also a major cause globally
- Results in undernutrition, poor growth, death
- Usually treated with oral rehydration solution, continuing to breast-feed
- Antibiotics only for cholera, dysentery, giardiasis, amoebiasis

Pneumonia
- Risk factors – low birthweight, young age, not breast-fed, vitamin A deficiency, overcrowding
- Predominantly bacterial
- Strategy to reduce mortality:
  - Prevention – breast feeding and hand hygiene
  - Prevention – immunisation
  - Treatment – effective case management by early diagnosis using WHO guidelines (fever, cough, tachypnoea, chest recession, head nodding) and prompt treatment with antibiotics

Measles
- Preventable by immunisation

Neonatal infection
- Remains major cause of death
- Mainly early-onset infection acquired at delivery

- Contact with animals, e.g. brucellosis.
- Increased susceptibility from immunodeficiency. This is usually secondary, e.g. post-autosplenectomy in sickle cell disease or splenectomy or nephrotic syndrome, resulting in increased susceptibility to encapsulated organisms (*Streptococcus pneumoniae*, *Haemophilus influenzae* and salmonella), or rarely, primary immune deficiency.

(iv) How ill is the child?

Red Flag features suggesting serious illness and the need for urgent investigation and treatment are:

- Fever >38°C if <3 months, >39°C if 3–6 months
- Colour – pale, mottled, blue
- Level of consciousness is reduced, neck stiffness, bulging fontanelle, status epilepticus, focal neurological signs or seizures
- Significant respiratory distress
- Bile-stained vomiting
- Severe dehydration or shock.

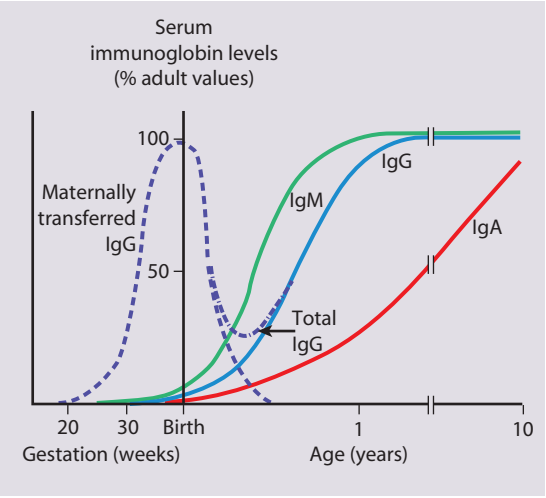


Figure 14.2 Serum immunoglobulin levels in the fetus and infant. When maternal immunoglobulin levels decline, infants become susceptible to viral infections.

Box 14.1 Septic screen

- Blood culture
- Full blood count including differential white cell count
- Acute phase reactant, e.g. C-reactive protein (CRP)
- Urine sample

Consider if indicated

- Chest X-ray
- Lumbar puncture (unless contraindicated)
- Rapid antigen screen on blood/CSF/urine
- Meningococcal and pneumococcal PCR on blood/CSF
- PCR for viruses in CSF (especially HSV and enterovirus).

(v) Is there a rash?

Rashes often accompany febrile illnesses. In some, the characteristics of the rash and other clinical features lead to a diagnosis, e.g. meningococcal septicaemia; in many, a specific diagnosis cannot be made clinically.

(vi) Is there a focus for infection?

Examination may identify a focus of infection (Fig. 14.3). If identified, investigations and management will be directed towards its treatment. However, if no focus is identified, this is often because it is the prodromal phase of a viral illness, but may indicate serious bacterial infection, especially urinary tract infection or septicaemia.

Management

Children who are not seriously ill can be managed at home with regular review by the parents, as long as they are given clear instructions (e.g. what clinical features should prompt reassessment by a doctor). Children who are significantly unwell, particularly if there is no focus of infection, will require investigations and observation or treatment in a paediatric assessment unit or A&E department or children's ward. A septic screen will be required (Box 14.1).

Parenteral antibiotics are given immediately to seriously unwell children, e.g. a third-generation cephalosporin such as cefotaxime or ceftriaxone if >3 months old. In infants 1–3 months old, cefotaxime (in case of septicaemia or meningitis) and ampicillin (in case of *Listeria* infection) are usually given. Aciclovir is given if herpes simplex encephalitis is suspected. Supportive care is given as indicated.

The child should not be underdressed. The use of antipyretic agents should be considered in children with fever who appear distressed or unwell. They should not be given if the child is otherwise well. Either paracetamol or ibuprofen can be used. They can be given alternatively if unresponsive to a single agent. Evidence that antipyretics prevent febrile seizures is lacking. There are NICE guidelines for the management of the child with fever.

Summary

The febrile child

- Upper respiratory tract infection (URTI) is an extremely common cause
- Check for otitis media
- Serious bacterial infection must be considered if there is no focus of infection, especially urinary tract infection or septicaemia, or there are Red Flag features of life-threatening illness
- The younger the child, the lower the threshold for performing a septic screen and starting antibiotics.

Serious life-threatening infections

Septicaemia

This is considered in Chapter 6 on Paediatric Emergencies.

Meningitis

Meningitis occurs when there is inflammation of the meninges covering the brain. This can be confirmed by finding inflammatory cells in the cerebrospinal fluid (CSF). Viral infections are the most common cause of meningitis, and most are self-resolving. Bacterial meningitis may have severe consequences. Other causes of non-infectious meningitis include malignancy and autoimmune diseases.

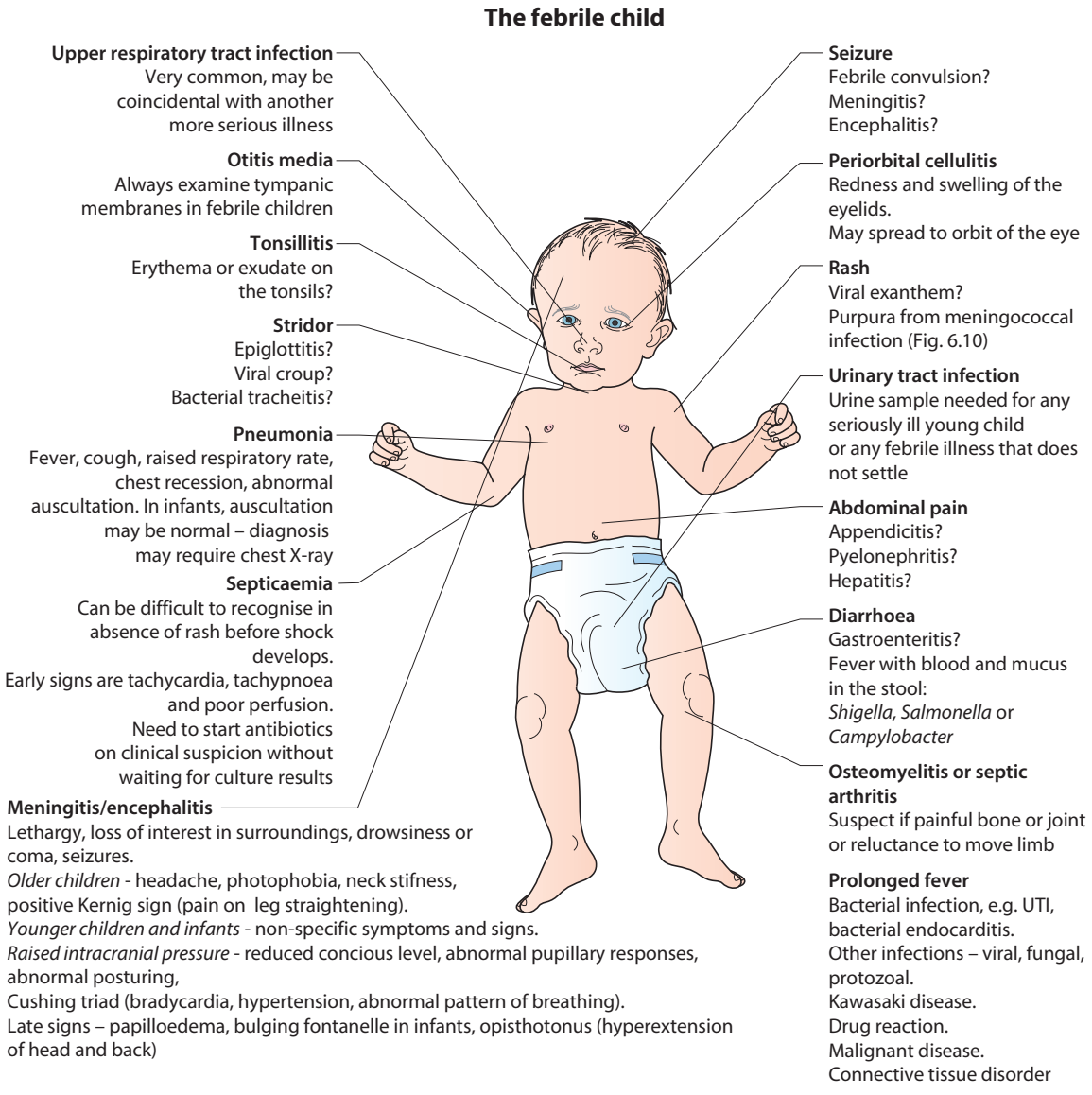


Figure 14.3 Some diagnostic clues to evaluating the febrile child.

Bacterial meningitis

Over 80% of patients with bacterial meningitis in the UK are younger than 16 years old. Bacterial meningitis remains a serious infection in children, with a 5–10% mortality. Over 10% of survivors are left with long-term neurological impairment.

Pathophysiology

Bacterial infection of the meninges usually follows bacteraemia. Much of the damage caused by meningeal infection results from the host response to infection and not from the organism itself. The release of inflammatory mediators and activated leucocytes, together with endothelial damage, leads to cerebral oedema, raised intracranial pressure and decreased cerebral blood flow. The inflammatory response below the meninges causes a vasculopathy resulting in cerebral cortical infarction, and fibrin deposits may block the resorption of CSF by the arachnoid villi, resulting in hydrocephalus.

Organisms

The organisms which commonly cause bacterial meningitis vary according to the child's age (Table 14.1).

Table 14.1 Organisms causing bacterial meningitis according to age

Neonatal–3 months	Group B streptococcus
	E. coli and other coliforms
	Listeria monocytogenes
1 month–6 years	*Neisseria meningitidis*
	Streptococcus pneumoniae
	Haemophilus influenzae
>6 years	*Neisseria meningitidis*
	Streptococcus pneumoniae

Presentation

The clinical features are listed in Figure 14.4. The early signs and symptoms of meningitis are non-specific, especially in infants and young children. Only children old enough to talk are likely to describe the classical meningitis symptoms of headache, neck stiffness and photophobia. But neck stiffness may also be seen in some children with tonsillitis and cervical lymphadenopathy. As children with meningitis may also be septicaemic, signs of shock, such as tachycardia, tachypnoea, prolonged capillary refill time, and hypotension, should be sought. Purpura in a febrile child of any age should be assumed to be due to meningococcal sepsis, even if the child does not appear unduly ill at the time; meningitis may or may not be present.

Investigations

The essential investigations are listed in Figure 14.4. A lumbar puncture is performed to obtain CSF to confirm the diagnosis, identify the organism responsible, and its antibiotic sensitivity. If any of the contraindications listed in Figure 14.4 are present, a lumbar puncture should not be performed, as under these circumstances, the procedure carries a risk of coning of the cerebellum through the foramen magnum. In these circumstances, a lumbar puncture can be postponed until the child's condition has stabilised. Even without a lumbar puncture, bacteriological diagnosis can be achieved in at least 50% of cases from the blood by culture or polymerase chain reaction (PCR), and rapid antigen screens can be performed on blood and urine samples. Throat swabs should also be obtained for bacterial and viral cultures. A serological diagnosis can be made on convalescent serum 4–6 weeks after the presenting illness if necessary.

Management

It is imperative that there is no delay in the administration of antibiotics and supportive therapy in a child with meningitis. The choice of antibiotics will depend on the likely pathogen. A third-generation cephalosporin, e.g. cefotaxime or ceftriaxone, is the preferred choice to cover the most common bacterial causes. Although still rare in the UK, pneumococcal resistance to penicillin and cephalosporins is increasing rapidly in certain parts of the world. The length of the course of antibiotics given depends on the causative organism and clinical response. Beyond the neonatal period, dexamethasone administered with the antibiotics reduces the risk of long-term complications such as deafness.

Cerebral complications

These include:

- *Hearing loss*. Inflammatory damage to the cochlear hair cells may lead to deafness. All children who have had meningitis should have an audiological assessment promptly, as children who become deaf may benefit from hearing amplification or a cochlear implant.
- *Local vasculitis*. This may lead to cranial nerve palsies or other focal lesions.
- *Local cerebral infarction*. This may result in focal or multifocal seizures, which may subsequently lead to epilepsy.
- *Subdural effusion*. Particularly associated with *Haemophilus influenzae* and pneumococcal meningitis. This is confirmed by CT scan. Most resolve spontaneously but may require prolonged antibiotic treatment.
- *Hydrocephalus*. May result from impaired resorption of CSF (communicating hydrocephalus) or blockage of the ventricular outlets by fibrin (non-communicating hydrocephalus). A ventricular shunt may be required.
- *Cerebral abscess*. The child's clinical condition deteriorates with the emergence of signs of a space-occupying lesion. The temperature will continue to fluctuate. It is confirmed on CT scan. Drainage of the abscess is required.

Prophylaxis

Prophylactic treatment with rifampicin to eradicate nasopharyngeal carriage is given to all household contacts for meningococcal meningitis and *Haemophilus influenzae* infection. It is not required for the patient if given a third-generation cephalosporin, as this will eradicate nasopharyngeal carriage. Household contacts of patients who have had group C meningococcal meningitis should be vaccinated with the meningococcal group C vaccine.

Partially treated bacterial meningitis

Children are frequently given oral antibiotics for a non-specific febrile illness. If they have early meningitis, this partial treatment with antibiotics may cause diagnostic problems. CSF examination shows a markedly raised number of white cells, but cultures are usually negative. Rapid antigen screens and PCR are helpful in these circumstances. Where the diagnosis is suspected clinically, a full course of antibiotics should be given.

Viral meningitis

Two-thirds of CNS infections are viral. Causes include enteroviruses, Epstein–Barr virus, adenoviruses and mumps. Mumps meningitis is now rare in the UK due to the MMR vaccine. Viral meningitis is usually much less severe than bacterial meningitis and a full recovery

Assessment & investigation of meningitis/encephalitis

History	Examination	Investigations
Fever	Fever	Full blood count and differential count
Headache	Purpuric rash (meningococcal disease)	Blood glucose and blood gas (for acidosis)
Photophobia	Neck stiffness (not always present in infants)	Coagulation screen, C-reactive protein
Lethargy	Bulging fontanelle in infants	Urea and electrolytes, liver function tests
Poor feeding/vomiting	Opisthotonus (arching of back)	Culture of blood, throat swab, urine, stool for bacteria and viruses
Irritability	Positive Brudzinski/Kernig signs	Rapid antigen test for meningitis organisms (can be done on blood, CSF, or urine)
Hypotonia	Signs of shock	Lumbar puncture for CSF unless contraindicated (see below for tests on CSF)
Drowsiness	Focal neurological signs	Serum for comparison of convalescent titres
Loss of consciousness	Altered conscious level	PCR of blood and CSF for possible organisms
Seizures	Papilloedema (rare)	If TB suspected: chest X-ray, Mantoux test, gastric washings or sputum, early morning urines
		Consider CT/MRI brain scan and EEG

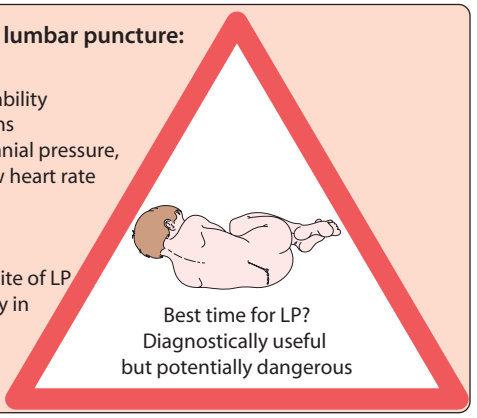
Signs associated with neck stiffness

Brudzinski sign – flexion of the neck with the child supine causes flexion of the knees and hips

Kernig sign – with the child lying supine and with the hips and knees flexed, there is back pain on extension of the knee

Contraindications to lumbar puncture:

- Cardiorespiratory instability
- Focal neurological signs
- Signs of raised intracranial pressure, e.g. coma, high BP, low heart rate or papilloedema
- Coagulopathy
- Thrombocytopenia
- Local infection at the site of LP
- If it causes undue delay in starting antibiotics



Typical changes in the CSF in meningitis or encephalitis, beyond the neonatal period

	Aetiology	Appearance	White blood cells	Protein	Glucose
Normal	—	Clear	0–5/mm^3	0.15–0.4 g/L	≥50% of blood
Meningitis	Bacterial	Turbid	Polymorphs:↑↑	↑↑	↓↓
	Viral	Clear	Lymphocytes:↑ (initially may be polymorphs)	Normal/↑	Normal/↓
	Tuberculosis	Turbid/clear/viscous	Lymphocytes:↑	↑↑↑	↓↓↓
Encephalitis	Viral/unknown	Clear	Normal/↑ lymphocytes	Normal/↑	Normal/↓

Figure 14.4 Assessment and investigation of meningitis and encephalitis.

can be anticipated. Diagnosis of viral meningitis can be confirmed by culture or PCR of CSF; culture of stool, urine, nasopharyngeal aspirate, throat swabs; and serology.

Uncommon pathogens and other causes

Where the clinical course is atypical or there is failure to respond to antibiotic and supportive therapy, unusual organisms, e.g. *Mycoplasma* or *Borrelia burgdorferi* (Lyme disease), or fungal infections need to be considered. Uncommon pathogens are particularly likely in children who are immunodeficient. Rarely, recurrent bacterial meningitis may occur in the immunodeficient or in children with structural abnormalities of the skull or meninges which facilitate bacterial access. Aseptic meningitis may be seen in malignancy or autoimmune disorders.

Neonatal meningitis

See Chapter 10.

Summary

Meningitis

- Predominantly a disease of infants and children
- Incidence has been reduced by immunisation
- Clinical features: non-specific in children under 18 months – fever, poor feeding, vomiting, irritability, lethargy, drowsiness, seizures or reduced consciousness; late signs – bulging fontanelle, neck stiffness and arched back (opisthotonos)
- Septicaemia can kill in hours; good outcome requires prompt resuscitation and antibiotics
- Any febrile child with a purpuric rash should be given intramuscular benzylpenicillin immediately and transferred urgently to hospital.

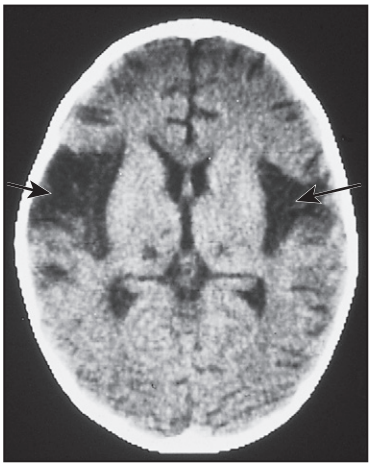


Figure 14.5 Herpes simplex encephalitis. The CT scan shows gross atrophy from loss of neural tissue in the temporoparietal regions (arrows).

Encephalitis/encephalopathy

Whereas in meningitis there is inflammation of the meninges, in encephalitis there is inflammation of the brain substance, although the meninges are often also affected. Encephalitis may be caused by:

- Direct invasion of the cerebrum by a neurotoxic virus (such as herpes simplex virus, HSV)
- Delayed brain swelling following a disordered neuroimmunological response to an antigen, usually a virus (post-infectious encephalopathy), e.g. following chickenpox
- A slow virus infection, such as HIV infection or subacute sclerosing panencephalitis (SSPE) following measles.

In encephalopathy from a non-infectious cause, such as a metabolic abnormality, the clinical features may be similar to an infectious encephalitis.

The clinical features and investigation of encephalitis are described in Figure 14.4. Most children present with fever, altered consciousness and often seizures. Initially, it may not be possible to clinically differentiate encephalitis from meningitis, and treatment for both should be started. The underlying causative organism is only detected in 50% of cases. In the UK, the most frequent causes of encephalitis are enteroviruses, respiratory viruses and herpesviruses (e.g. herpes simplex virus, varicella and HHV6). Worldwide, microorganisms causing encephalitis include *Mycoplasma, Borrelia burgdorferi* (Lyme disease), *Bartonella henselae* (cat scratch disease), rickettsial infections (e.g. Rocky Mountain spotted fever) and the arboviruses.

Herpes simplex virus (HSV) is a rare cause of childhood encephalitis but it may have devastating long-term consequences. All children with encephalitis should therefore be treated initially with high-dose intravenous aciclovir, since this is a very safe treatment. Most affected children do not have outward signs of herpes infection, such as cold sores, gingivostomatitis or skin lesions. The PCR of the CSF may be positive for HSV. As HSV encephalitis is a destructive infection, the EEG and CT/MRI scan may show focal changes, particularly within the temporal lobes (Fig. 14.5). These tests may be normal initially and need to be repeated after a few days if the child is not improving. Later confirmation of the diagnosis may be made from HSV antibody production in the CSF. Proven cases of HSV encephalitis or cases where there is a high index of suspicion should be treated with intravenous aciclovir for 3 weeks, as relapses may occur after shorter courses. Untreated, the mortality rate from HSV encephalitis is over 70% and survivors usually have severe neurological sequelae.

Summary

Encephalitis

- Onset can be insidious and includes behavioural change
- Consider if HSV (herpes simplex virus) could be the cause
- Treat potential HSV with parenteral high-dose aciclovir until diagnosis is excluded.

Toxic shock syndrome

Toxin-producing *Staphylococcus aureus* and group A streptococci can cause this syndrome, which is characterised by:

- Fever >39°C
- Hypotension
- Diffuse erythematous, macular rash.

The toxin can be released from infection at any site, including small abrasions or burns, which may look minor. The toxin acts as a superantigen and, in addition to the features above, causes organ dysfunction, including:

- Mucositis (Fig. 14.6): conjunctivae, oral mucosa, genital mucosa

- Gastrointestinal: vomiting/diarrhoea
- Renal impairment
- Liver impairment
- Clotting abnormalities and thrombocytopenia
- Central nervous system: altered consciousness.

Intensive care support is required to manage the shock. Areas of infection should be surgically debrided. Antibiotics often include a third-generation cephalosporin (such as ceftriaxone) together with clindamycin, which acts on the bacterial ribosome to switch off toxin production. Intravenous immunoglobulin may be given to neutralise circulating toxin. About 1–2 weeks after the onset of the illness, there is desquamation of the palms, soles, fingers and toes.

PVL-producing *Staphylococcus aureus* causes recurrent skin and soft tissue infections, but can also cause necrotising fasciitis and a necrotising haemorrhagic pneumonia following an influenza-like illness; they carry a high mortality. PVL-producing *Staphylococcus aureus* produces a toxin called Panton-Valentine leukocidin (PVL) which has emerged in the UK and other countries. PVL is produced by fewer than 2% of *Staphylococcus aureus* strains (both methicillin-sensitive *Staphylococcus aureus* (MSSA) and methicillin-resistant *Staphylococcus aureus* (MRSA)). In children, the procoagulant state frequently results in venous thrombosis.

Necrotising fasciitis/cellulitis

This is a severe subcutaneous infection, often involving tissue planes from the skin down to fascia and muscle. The area involved may enlarge rapidly, leaving poorly perfused necrotic areas of tissue, usually at the centre. There is severe pain and systemic illness, which may require intensive care. The invading organism may be *Staphylococcus aureus* or a group A streptococcus, with or without another synergistic anaerobic organism. Intravenous antibiotic therapy alone is not sufficient to treat this condition. Without surgical intervention and debridement of necrotic tissue, the infection will continue to spread. Clinical suspicion of necrotising fasciitis warrants urgent surgical consultation and intervention. Intravenous immunoglobulin (IVIG) may also be given.



Figure 14.6 A child with toxic shock syndrome receiving intensive care, including artificial ventilation via a nasotracheal tube. The lips are red and the eyelids are oedematous from capillary leak. (Courtesy of Professor Mike Levin.)

Specific bacterial infections

Meningococcal infection

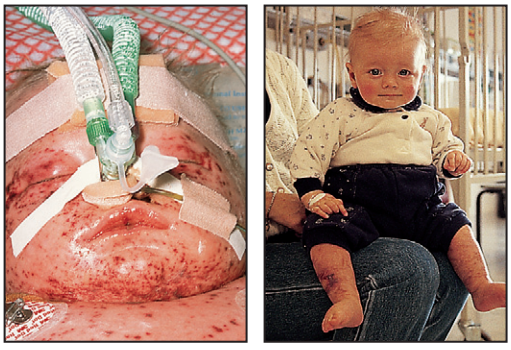
Meningococcal infection is a disease that strikes fear into both parents and doctors, as it can kill previously healthy children within hours (Case History 14.1). However, of the three main causes of bacterial meningitis, meningococcal has the lowest risk of long-term neurological sequelae, with most survivors recovering fully. The septicaemia is usually accompanied by a purpuric rash which may start anywhere on the body and then spread. The rash may or may not be present with meningococcal meningitis. Characteristic lesions are non-blanching on palpation, irregular in size and outline and have a necrotic centre (Fig. 14.8a,b). Any febrile child who develops a purpuric rash should be treated immediately, at home or in the general practitioner's surgery, with systemic antibiotics such as penicillin before urgent admission to hospital. Although there are now polysaccharide conjugate vaccines

Case History

14.1 Meningococcal septicaemia

This 7-month-old boy presented with a 12-hour history of lethargy and a spreading purpuric rash. In hospital, he required immediate resuscitation and transfer to a paediatric intensive care unit for multi-organ failure (Fig. 14.7a). The gross oedema is from leak of capillary fluid into the tissues. He required colloid and inotropic support and peritoneal dialysis for renal failure. He made a full recovery (Fig. 14.7b).

Meningococcal septicaemia can kill children in hours. Optimal outcome requires immediate recognition, prompt resuscitation and antibiotics.



(a) (b)

Figure 14.7 **(a)** A boy with meningococcal septicaemia receiving intensive care. **(b)** After full recovery. (Courtesy of Dr Parviz Habibi.)

Meningococcal septicaemia



(a)



(b)

Figure 14.8 Rash of meningococcal infection. **(a)** Characteristic purpuric skin lesions, irregular in size and outline and with a necrotic centre. **(b)** The lesions may be extensive, when it is called '*purpura fulminans*'.

against groups A and C meningococcus, there is still no effective vaccine for group B meningococcus, which accounts for the majority of isolates in the UK.

Any febrile child with a purpuric rash should be given intramuscular benzylpenicillin immediately and transferred urgently to hospital.

Pneumococcal infections

Streptococcus pneumoniae is often carried in the nasopharynx of healthy children. Asymptomatic carriage is particularly prevalent among young children and may be responsible for the transmission of pneumococcal disease to other individuals by respiratory droplets. The organism may cause pharyngitis, otitis media, conjunctivitis, sinusitis as well as 'invasive' disease (pneumonia, bacterial sepsis and meningitis). Invasive disease, which carries a high burden of morbidity and mortality, mainly occurs in young infants as their immune system responds poorly to encapsulated pathogens such as pneumococcus. With the inclusion of the 13-valent pneumococcal vaccine into the standard immunisation schedule in the UK, the incidence of invasive disease has declined. Children at increased risk, e.g. from hyposplenism, should also be given daily prophylactic penicillin to prevent infection by strains not covered by the vaccine.

Haemophilus infection

H. influenzae type b was an important cause of systemic illness in children, including otitis media, pneumonia, epiglottitis, cellulitis, osteomyelitis and septic arthritis and was the second most common cause of meningitis in the UK. Immunisation has been highly effective and it now rarely causes systemic disease.

Staphylococcal and group A streptococcal infections

Staphylococcal and streptococcal infections are usually caused by direct invasion of the organisms. They may also cause disease by releasing toxins which act as superantigens. Whereas conventional antigens stimulate only a small subset of T cells which have a specific receptor, superantigens bind to a part of the T-cell receptor which is shared by many T cells and therefore stimulates massive T-cell proliferation and cytokine release. Other diseases following staphylococcal and streptococcal infections are immune-mediated.



Figure 14.9 Impetigo showing characteristic confluent honey-coloured crusted lesions. (Courtesy of Dr Paul Hutchins.)

Summary

Pneumococcal infection

- Causes not only minor infections such as otitis media but also invasive disease
- Susceptibility is increased in hyposplenism (e.g. sickle cell disease and nephrotic syndrome)
- Vaccination is included in the standard immunisation schedule.

Impetigo

This is a localised, highly contagious, staphylococcal and/or streptococcal skin infection, most common in infants and young children. It is more common where there is pre-existing skin disease, e.g. atopic eczema. Lesions are usually on the face, neck and hands and begin as erythematous macules which may become vesicular/pustular or even bullous (Fig. 14.9). Rupture



Figure 14.10 Periorbital cellulitis. It should be treated promptly with intravenous antibiotics to prevent spread into the orbit.

of the vesicles with exudation of fluid leads to the characteristic confluent honey-coloured crusted lesions. Infection is readily spread to adjacent areas and other parts of the body by autoinoculation of the infected exudate. Topical antibiotics (e.g. mupirocin) are sometimes effective for mild cases. Narrow-spectrum systemic antibiotics (e.g. flucloxacillin) are needed for more severe infections, although more broad-spectrum antibiotics such as co-amoxiclav or cefaclor have simpler oral administration regimens, taste better and therefore have better adherence. Affected children should not go to nursery or school until the lesions are dry. Nasal carriage is an important source of infection which can be eradicated with a nasal cream containing mupirocin or chlorhexidine and neomycin.

Boils

These are infections of hair follicles or sweat glands, usually caused by *Staphylococcus aureus*. Treatment is with systemic antibiotics and occasionally surgery. Recurrent boils are usually from persistent nasal carriage in the child or family acting as a reservoir for reinfection. Only rarely are they a manifestation of immune deficiency.

Periorbital cellulitis

In periorbital cellulitis there is fever with erythema, tenderness and oedema of the eyelid (Fig. 14.10). It is almost always unilateral. In young, unimmunised children it may also be caused by *Haemophilus influenzae* type b which may also be accompanied by infection at other sites, e.g. meningitis. It may follow local trauma to the skin. In older children, it may spread from a paranasal sinus infection or dental abscess. Periorbital cellulitis should be treated promptly with intravenous antibiotics to prevent posterior spread of the infection to become an orbital cellulitis. In orbital cellulitis, there is proptosis, painful or limited ocular movement and reduced visual acuity. It may be complicated by abscess formation, meningitis or cavernous sinus thrombosis. Where orbital cellulitis is suspected, a CT scan should be performed to assess the posterior spread of infection and a lumbar puncture may be required to exclude meningitis.



Figure 14.11 Staphylococcal scalded skin syndrome. Its appearance must not be mistaken for a scald from non-accidental injury.

Scalded skin syndrome

This is caused by an exfoliative staphylococcal toxin which causes separation of the epidermal skin through the granular cell layers. It affects infants and young children, who develop fever and malaise and may have a purulent, crusting, localised infection around the eyes, nose and mouth with subsequent widespread erythema and tenderness of the skin. Areas of epidermis separate on gentle pressure (Nikolsky sign), leaving denuded areas of skin (Fig. 14.11), which subsequently dry and heal without scarring. Management is with an intravenous anti-staphylococcal antibiotic, analgesia and monitoring of fluid balance.

Summary

Staphylococcal and streptococcal infections

- Symptoms are caused by direct invasion of bacteria or by release of toxins
- Impetigo is highly contagious
- Periorbital cellulitis should be treated aggressively to prevent spread to the orbit or brain
- Scalded skin syndrome is rare but serious.

Common viral infections

Many of the common childhood infections present with fever and a rash (Table 14.2). Incubation periods vary from 24 h for viral gastroenteritis, to about 2 weeks for chickenpox, but for some diseases, such as HIV, the length of time between exposure and the development of symptomatic illness may extend to many years. This is a reflection of host–pathogen interactions; an effective initial host response may result in a prolonged period of clinical latency, whereas an ineffective response permits rapid evolution of disease.

The infectious period characteristically begins a day or two before the rash appears and, for purposes of nursery/school exclusion, is generally considered to last until the rash has resolved or the lesions have dried up. For details about incubation and exclusion periods, see the Health Protection Agency website (http://www.hpa.org.uk).

Table 14.2 Causes of fever and a rash

Maculopapular rash	
Viral	HHV6 or 7 (Roseola infantum) – <2 years old
	Enteroviral rash
	Parvovirus ('slapped cheek') – usually school-age
	Measles – uncommon if immunised
	Rubella – uncommon if immunised
Bacterial	Scarlet fever (group A streptococcus)
	Erythema marginatum – rheumatic fever
	Salmonella typhi (typhoid fever) – classically rose spots
	Lyme disease – erythema migrans
Other	Kawasaki disease
	Juvenile idiopathic arthritis
Vesicular, bullous, pustular	
Viral	Varicella-zoster virus – chickenpox, shingles
	Herpes simplex virus
	Coxsackie – hand, foot and mouth
Bacterial	Impetigo – characteristic crusting
	Boils – infection of hair follicles/sweat glands
	Staphylococcal bullous impetigo
	Staphylococcal scalded skin
	Toxic epidermal necrolysis
Other	Erythema multiforme; Stevens–Johnson syndrome
Petechial, purpuric	
Bacterial	Meningococcal, other bacterial sepsis
	Infective endocarditis
Viral	Enterovirus and other viral infections
Other	Henoch–Schönlein purpura (HSP)
	Thrombocytopenia
	Vasculitis
	Malaria

The human herpesviruses

There are currently eight known human herpesviruses: herpes simplex virus 1 and 2 (HSV1 and HSV2), varicella zoster virus (VZV), cytomegalovirus (CMV), Epstein–Barr virus (EBV), and human herpesviruses 6, 7 and 8 (HHV 6–8). HHV8 is associated with Kaposi sarcoma in HIV-coinfected individuals. The other herpesviruses will be discussed in this section, in order of their prevalence.

The hallmark of the herpesviruses is that, after primary infection, latency is established and there is long-term persistence of the virus within the host, usually in a dormant state. After certain stimuli, reactivation of infection may occur.

Herpes simplex infections

Herpes simplex virus (HSV) usually enters the body through the mucous membranes or skin, and the site of the primary infection may be associated with intense local mucosal damage. HSV1 is usually associated with lip and skin lesions, and HSV2 with genital lesions, but both viruses can cause both types of disease. The wide variety of clinical manifestations are described below. Treatment is with aciclovir, a viral DNA polymerase inhibitor, which may be used to treat severe symptomatic skin, ophthalmic, cerebral and systemic infections.



Figure 14.12 Vesicles with ulceration in gingivostomatitis.

Asymptomatic

Herpes simplex infections are very common and are mostly asymptomatic.

Gingivostomatitis

This is the most common form of primary HSV illness in children. It usually occurs from 10 months to 3 years of age. There are vesicular lesions on the lips, gums and anterior surfaces of the tongue and hard palate, which often progress to extensive, painful ulceration with bleeding (Fig. 14.12). There is a high fever and the child



Figure 14.13 Eczema herpeticum.

is very miserable. The illness may persist for up to 2 weeks. Eating and drinking are painful, which may lead to dehydration. Management is symptomatic, but severe disease may necessitate intravenous fluids and aciclovir.

Skin manifestations

Mucocutaneous junctions and damaged skin are particularly prone to infection. 'Cold sores' are recurrent HSV1 lesions on the gingival (lip) margin.

Eczema herpeticum – In this serious condition, widespread vesicular lesions develop on eczematous skin (Fig. 14.13). This may be complicated by secondary bacterial infection, which may result in septicaemia.

Herpetic whitlows – These are painful, erythematous, oedematous white pustules on the site of broken skin on the fingers. Spread is by auto-inoculation from gingivostomatitis and infected adults kissing their children's fingers. In sexually active adolescents, HSV2 may be the cause.

Eye disease

Eye disease may cause a blepharitis or conjunctivitis. It may extend to involve the cornea, producing dendritic ulceration. This can lead to corneal scarring and ultimately loss of vision. Any child with herpetic lesions near or involving the eye requires ophthalmic investigation of the cornea by slit lamp examination.

Central nervous system infection

Neonatal infection (see Ch. 10) – The infection may be focal, affecting the skin or eyes or encephalitis or may be widely disseminated. Its morbidity and mortality are high.

Infection in the immunocompromised host – Infection may be severe. Cutaneous lesions may spread to involve adjacent sites, e.g. oesophagitis and proctitis. Pneumonia and disseminated infections involving multiple organs are serious complications.

Chickenpox (primary varicella zoster infection)

Clinical features

These are shown in Figure 14.14.

There are a number of rare but serious complications that can occur in previously healthy children:

- *Secondary bacterial infection* with staphylococci, group A streptococcal, or other organisms. May lead to further complications such as toxic shock syndrome or necrotising fasciitis. Secondary bacterial infection should be considered where there is onset of a new fever or persistent high fever after the first few days.
- *Encephalitis*. This may be generalised, usually occurring early during the illness. In contrast to the encephalitis caused by HSV, the prognosis is good. Most characteristic is a VZV-associated cerebellitis. This usually occurs about a week after the onset of rash. The child is ataxic with cerebellar signs. It usually resolves within a month.
- *Purpura fulminans*. This is the consequence of vasculitis in the skin and subcutaneous tissues. It is best known in relation to meningococcal disease and can lead to loss of large areas of skin by necrosis. It may rarely occur after VZV infection due to production of antiviral antibodies which cross-react and inactivate the coagulation factor protein S. There is subsequent dysregulation of fibrinolysis and an increased risk of clotting, most often manifest in the skin.

In the immunocompromised, primary varicella infection may result in severe progressive disseminated disease, which has a mortality of up to 20%. The vesicular eruptions persist and may become haemorrhagic. The disease in the neonatal period is described in Chapter 10.

Watch for the child with chickenpox whose fever initially settles, but then recurs a few days later – it is likely to be due to secondary bacterial infection.

Treatment and prevention

Oral aciclovir has minimal benefit and is not recommended in the UK. Immunocompromised children should be treated with intravenous aciclovir initially.

Summary

Herpes simplex virus infections

- Most are asymptomatic
- Gingivostomatitis – may necessitate intravenous fluids and aciclovir
- Skin manifestations – mucocutaneous junctions, e.g. lips and damaged skin
- Eczema herpeticum – may result in secondary bacterial infection and septicaemia
- Herpetic whitlows – painful pustules on the fingers
- Eye disease – blepharitis, conjunctivitis, corneal ulceration and scarring
- CNS – aseptic meningitis, encephalitis
- Pneumonia and disseminated infection in the immunocompromised.

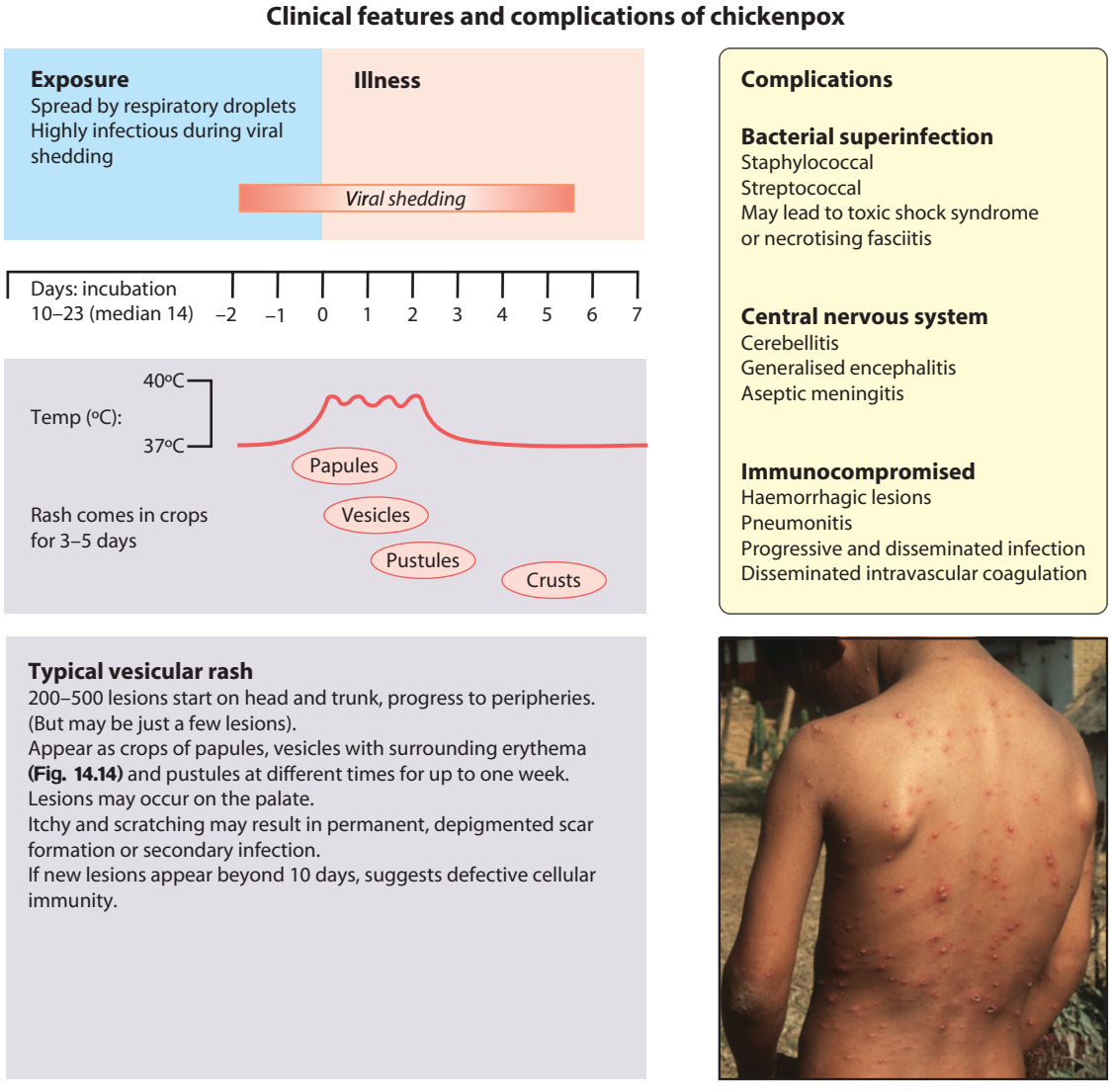


Figure 14.14 Clinical features and complications of chickenpox.

Oral valaciclovir can be substituted if organ dissemination has not occurred. Valaciclovir can be considered for adolescents and adults unlucky enough to develop primary VZV infection, as it is more severe when contracted beyond childhood. Human varicella zoster immunoglobulin (VZIG) is recommended for high-risk immunosuppressed individuals with deficient T-lymphocyte function following contact with chickenpox. Protection from infection with zoster immunoglobulin is not absolute, and depends on how soon after contact with chickenpox it is given.

Beware of admitting a chickenpox contact to a clinical area with immunocompromised children, in whom it can disseminate and be fatal.

Shingles (herpes zoster)

Shingles is uncommon in children. It is caused by reactivation of latent varicella-zoster virus (VZV), causing a vesicular eruption in the dermatomal distribution of sensory nerves (shingles). It occurs most commonly in the thoracic region, although any dermatome can be affected (Fig. 14.15). Children, unlike adults, rarely suffer neuralgic pain with shingles. Shingles in childhood is more common in those who had primary infection in the first year of life. Recurrent or multidermatomal shingles is strongly associated with underlying immune suppression, e.g. HIV infection. In the immunocompromised, reactivated infection can also disseminate to cause severe disease.

Recurrent or multidermatomal shingles suggests a T-cell immune defect.

Summary

Chickenpox

- Clinical features – fever and itchy, vesicular rash which crops for up to 7 days
- Complications – secondary bacterial infection, encephalitis; disseminated disease in the immunocompromised
- Human varicella zoster immunoglobulin (VZIG) – if immunosuppressed and in contact with chickenpox or if maternal chickenpox shortly before or after delivery
- Treatment is symptomatic; i.v. aciclovir for severe chickenpox or the immunocompromised.



Figure 14.15 Herpes zoster (shingles) in a child. Distribution is along the S1 dermatome. (Courtesy of Dr Sam Walters.)

Epstein–Barr virus: infectious mononucleosis (glandular fever)

Epstein–Barr virus (EBV) is the major cause of the infectious mononucleosis syndrome, but it is also involved in the pathogenesis of Burkitt lymphoma, lymphoproliferative disease in immunocompromised hosts and nasopharyngeal carcinoma. The virus has a particular tropism for B lymphocytes and epithelial cells of the pharynx. Transmission usually occurs by oral contact and the majority of infections are subclinical.

Older children, and occasionally young children, may develop a syndrome with:

- fever
- malaise
- tonsillopharyngitis – often severe, limiting oral ingestion of fluids and food; rarely, breathing may be compromised
- lymphadenopathy – prominent cervical lymph nodes, often with diffuse adenopathy.

Other features include:

- petechiae on the soft palate
- splenomegaly (50%), hepatomegaly (10%)
- a maculopapular rash (5%)
- jaundice.

Diagnosis is supported by:

- atypical lymphocytes (numerous large T cells seen on blood film)
- a positive Monospot test (the presence of heterophile antibodies, i.e. antibodies that agglutinate sheep or horse erythrocytes but which are not absorbed by guinea pig kidney extracts – this test is often negative in young children with the disease)
- seroconversion with production of IgM and IgG to Epstein–Barr virus antigens.

Symptoms may persist for 1–3 months but ultimately resolve. They are caused by the host immune response to the infection, rather than the virus itself.

Treatment is symptomatic. When the airway is severely compromised, corticosteroids may be considered. In 5% of infected individuals, group A streptococcus is grown from the tonsils. This may be treated with penicillin. Ampicillin or amoxicillin may cause a florid maculopapular rash in children infected with EBV and should be avoided.

Cytomegalovirus

Cytomegalovirus (CMV) is usually transmitted via saliva, genital secretions or breast milk, and more rarely via blood products, organ transplants and transplacentally. The virus causes mild or subclinical infection in normal hosts. In developed countries, about half of the adult population show serological evidence of past infection. In developing countries, most children have been infected by 2 years of age, often via breast milk. In the immunocompromised and the fetus, CMV is an important pathogen.

As with EBV, CMV may cause a mononucleosis syndrome. Pharyngitis and lymphadenopathy are not usually as prominent as in EBV infections. Patients may have atypical lymphocytes on the blood film but are heterophile antibody-negative. Maternal CMV infection may result in congenital infection (see Ch. 9), which may be present at birth or develop when older. In the immunocompromised host, CMV can cause retinitis, pneumonitis, bone marrow failure, encephalitis, hepatitis, colitis and oesophagitis. It is a very important pathogen following organ transplantation. Organ recipients are closely monitored for evidence of CMV activation by sensitive tests such as blood polymerase chain reaction (PCR). Interventions used to reduce the risk of transmission of CMV disease are CMV-negative blood for transfusions and anti-CMV drug prophylaxis; also, if possible, CMV-positive organs are not transplanted into CMV-negative recipients.

CMV disease may be treated with ganciclovir or foscarnet, but both have serious side-effects.

Human herpesvirus 6 (HHV6) and HHV7

Human herpesvirus 6 (HHV6) and HHV7 are closely related and have similar presentations, although HHV6 is more prevalent. Most children are infected with HHV6 or HHV7 by the age of 2 years, usually from the oral secretions of a family member. They classically cause exanthem subitum (also known as roseola infantum), characterised by a high fever with malaise lasting a few days, followed by a generalised macular rash, which appears as the fever wanes. Many children have a febrile illness without rash, and many have a subclinical infection. Exanthem subitum is frequently clinically misdiagnosed as measles or rubella; these infections are rare in the UK and if suspected should be confirmed serologically. Another frequent occurrence in primary HHV6 infection is that infants seen by a doctor during the febrile stage are prescribed antibiotics, and when the rash appears, it is erroneously attributed to an 'allergic' reaction to the drug. Primary HHV6/HHV7 infections are a common cause of febrile convulsions. Rarely, they may cause aseptic meningitis, encephalitis, hepatitis, or an infectious mononucleosis-like syndrome.

Parvovirus B19

Parvovirus B19 causes erythema infectiosum or fifth disease (so-named because it was the fifth disease to be described of a group of illnesses with similar rashes), also called slapped-cheek syndrome. Infections can occur at any time of the year, although outbreaks are most common during the spring months. Transmission is via respiratory secretions from viraemic patients, by vertical transmission from mother to fetus and by transfusion of contaminated blood products. Parvovirus B19 infects the erythroblastoid red cell precursors in the bone marrow.

Parvovirus causes a range of clinical syndromes:

- Asymptomatic infection – common; about 5–10% of preschool children and 65% of adults have antibodies
- Erythema infectiosum – the most common illness, with a viraemic phase of fever, malaise, headache and myalgia followed by a characteristic rash a week later on the face ('slapped-cheek'), progressing to a maculopapular, 'lace'-like rash on the trunk and limbs; complications are rare in children, although arthralgia or arthritis is common in adults
- Aplastic crisis – the most serious consequence of parvovirus infection; it occurs in children with chronic haemolytic anaemias, where there is an increased rate of red cell turnover (e.g. sickle cell disease or thalassaemia); and in immunodeficient children (e.g. malignancy) who are unable to produce an antibody response to neutralise the infection
- Fetal disease – transmission of maternal parvovirus infection may lead to fetal hydrops and death due to severe anaemia, although the majority of infected fetuses will recover.

Summary

Parvovirus

- Usually asymptomatic or erythema infectiosum
- Can cause aplastic crisis in haemolytic anaemias (e.g. sickle cell) or the fetus (causes hydrops).

Enteroviruses

Human enteroviruses, of which there are many (including the coxsackie viruses, echoviruses and polioviruses), are a common cause of childhood infection. Transmission is primarily by the faecal–oral route. Following replication in the pharynx and gut, the virus spreads to infect other organs. Infections occur most commonly in the summer and autumn. Over 90% of infections are asymptomatic or cause a non-specific febrile illness, sometimes with a rash usually over the trunk that is blanching or consists of fine petechiae. A history of loose stools or some vomiting, or a contact history, would be supportive. The child is not usually systemically unwell, but if the rash is non-blanching, admission for observation and 48 h of parenteral antibiotics (such as ceftriaxone) is indicated. It is better to treat a number of enteroviral infections than to send home a child with meningococcal disease, only to have them return moribund 12 h later.

Other characteristic clinical syndromes exist and are listed below. (For polioviruses, see the Immunisation section, below.)

Hand, foot and mouth disease

Painful vesicular lesions on the hands, feet, mouth and tongue, and often on the buttocks. Systemic features are mild. The disease subsides within a few days, with fluids and analgesia.

Herpangina

Vesicular and ulcerated lesions on the soft palate and uvula causing anorexia, pain on swallowing and fever. Resolves with fluids and analgesia.

Meningitis/encephalitis

Aseptic meningitis is caused by many of the enteroviruses. Complete recovery can be expected.

Pleurodynia (Bornholm disease)

An acute illness with fever, pleuritic chest pain and muscle tenderness. There may be a pleural rub but examination is otherwise normal. Recovery is within a few days.

Myocarditis and pericarditis

Rare. Heart failure associated with a febrile illness and ECG evidence of myocarditis.

Summary

Enterovirus infection

- Mostly asymptomatic or self-limiting illness with rash, which may be purpuric
- Can cause hand, foot and mouth disease, herpangina, or meningitis/encephalitis.

Uncommon viral infections

Measles

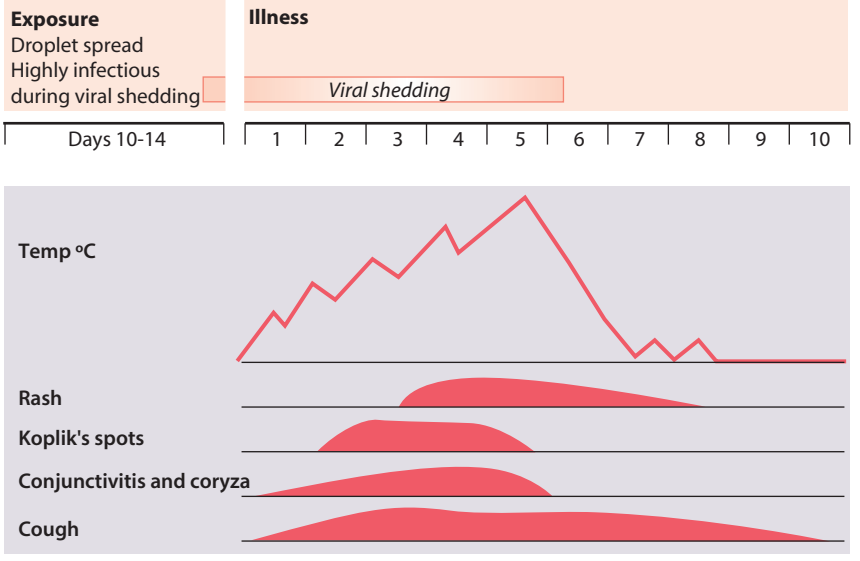
Health practitioners in the UK need to be aware of measles due to the rise in cases following public anxiety about the MMR vaccination (see the Immunisation section, below), as well as it continuing to be a major cause of morbidity and death worldwide. As with chickenpox and parvovirus, older children and adults tend to have more severe disease than the very young. For epidemiological tracking of infection, virological or serological confirmation of clinical cases of measles should be undertaken by testing either blood or saliva.

Clinical features

These are shown in Figure 14.16. There are a number of serious complications which can occur in previously healthy children:

- *Encephalitis* occurs in about 1 in 5000, about 8 days after the onset of the illness. Initial symptoms are headache, lethargy and irritability, proceeding to convulsions and ultimately coma. Mortality is 15%. Serious long-term sequelae include seizures, deafness, hemiplegia and severe learning difficulties, affecting up to 40% of survivors.
- *Subacute sclerosing panencephalitis* (*SSPE*) is a rare but devastating illness manifesting, on average, 7 years after measles infection in about 1 in 100 000 cases. Most children who develop SSPE had primary measles infection before 2 years of age. SSPE is caused by a variant of the measles virus which persists in the central nervous system. The disorder presents with loss of neurological function, which progresses over several years to dementia and death. The diagnosis is essentially

Clinical features and complications of measles



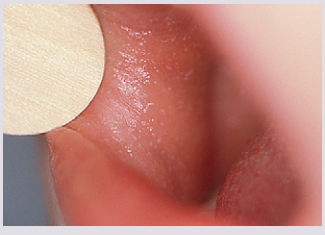
Complications

Respiratory
Pneumonia
Secondary bacterial infection and otitis media
Tracheitis

Neurological
Febrile convulsions
EEG abnormalities
Encephalitis
Subacute sclerosing panencephalitis (SSPE)

Other
Diarrhoea
Hepatitis
Appendicitis
Corneal ulceration
Myocarditis

Koplik's spots
White spots on buccal mucosa, seen against bright red background. Pathognomonic, but difficult to see.



Rash
Spreads downwards, from behind the ears to the whole of the body. Discrete, maculopapular rash initially, becomes blotchy and confluent. May desquamate in the second week.



Figure 14.16 Clinical features and complications of measles.

clinical, supported by finding high levels of measles antibody in both blood and cerebrospinal fluid and by characteristic EEG abnormalities. Since the introduction of measles immunisation, it has become extremely rare.

In developing countries, where malnutrition and vitamin A deficiency lead to impaired cell-mediated immunity, measles often follows a protracted course with severe complications. Impaired cellular immune responses such as in HIV infection may result in a modified or absent rash, with an increased risk of dissemination, including giant-cell pneumonia or encephalitis.

Treatment

Treatment for measles is symptomatic. Children who are admitted to hospital should be isolated. In immunocompromised patients, the antiviral drug ribavirin may be used. Vitamin A, which may modulate the immune response, should be given in developing countries.

Prevention

Prevention by immunisation is the most successful strategy for reducing the morbidity and mortality of measles.

Measles remains a major cause of death in childhood in developing countries.

Summary

Measles

- Incidence has declined dramatically since immunisation was introduced; a recent small increase has resulted from the fall in immunisation uptake
- Clinical features: fever, cough, runny nose, conjunctivitis, marked malaise, Koplik spots, maculopapular rash
- Complications: common if malnourished or immunocompromised; major cause of death in developing countries.

Mumps

Mumps occurs worldwide, but its incidence has declined dramatically because of the mumps component of the MMR vaccine. Following the decrease in the uptake of the MMR immunisation in the late 1990s, there has been a rise in unimmunised children and unvaccinated young adults. Mumps usually occurs in the winter and spring months. It is spread by droplet infection to the respiratory tract where the virus replicates within epithelial cells. The virus gains access to the parotid glands before further dissemination to other tissues.

Clinical features

The incubation period is 15–24 days. Onset of the illness is with fever, malaise and parotitis, but in up to 30% of cases, the infection is subclinical. Only one side may be swollen initially, but bilateral involvement usually occurs over the next few days. The parotitis is uncomfortable and children may complain of earache or pain on eating or drinking. Examination of the parotid duct may show redness and swelling. Occasionally, parotid swelling may be absent. The fever usually disappears within 3–4 days. Plasma amylase levels are often elevated and, when associated with abdominal pain, may be evidence of pancreatic involvement. Infectivity is for up to 7 days after the onset of parotid swelling. The illness is generally mild and self-limiting. Although hearing loss can follow mumps, it is usually unilateral and transient.

Viral meningitis and encephalitis

Lymphocytes are seen in the CSF in about 50%, meningeal signs are only seen in 10%, and encephalitis in about 1 in 5000. The common clinical features are headache, photophobia, vomiting and neck stiffness.

Orchitis

This is the most feared complication, although it is uncommon in prepubertal males. When it does occur, it is usually unilateral. Although there is some evidence of a reduction in sperm count, infertility is actually extremely unusual. Rarely, oophoritis, mastitis and arthritis may occur.

Rubella (German measles)

Rubella is generally a mild disease in childhood. It occurs in winter and spring. It is an important infection, as it can cause severe damage to the fetus (see Ch. 9). The incubation period is 15–20 days. It is spread by the respiratory route, frequently from a known contact. The prodrome is usually mild with a low-grade fever or none at all. The maculopapular rash is often the first sign of infection, appearing initially on the face and then spreading centrifugally to cover the whole body. It fades in 3–5 days. Unlike in adults, the rash is not itchy. Lymphadenopathy, particularly the suboccipital and postauricular nodes, is prominent. Complications are rare in childhood but include arthritis, encephalitis, thrombocytopenia and myocarditis. Clinical differentiation from other viral infections is unreliable. The diagnosis should be confirmed serologically if there is any risk of exposure of a non-immune pregnant woman. There is no effective antiviral treatment. Prevention therefore lies in immunisation.

Summary

Rubella

Importance – congenital infection.

Prolonged fever

Most childhood infections are acute and resolve in a few days. If not, the child needs to be reassessed for complications of the original illness, e.g. a secondary bacterial infection, or the source of infection may not have been identified, e.g. urinary tract infection. Often, the child has developed another unrelated febrile illness. Assessment of prolonged fever also needs to be made for prompt recognition of Kawasaki disease to avoid complications. Causes of prolonged fever are listed in Box 14.2.

Kawasaki disease

Kawasaki disease (KD) is a systemic vasculitis. Although uncommon, it is an important diagnosis to make because aneurysms of the coronary arteries are a potentially devastating complication. Prompt treatment reduces their incidence.

Kawasaki disease mainly affects children of 6 months to 4 years old, with a peak at the end of the first year. The disease is more common in children of Japanese and, to a lesser extent, Afro-Caribbean ethnicity, than in Caucasians. Young infants tend to be more severely affected than older children and are more likely to have 'incomplete' cases, in which not all the cardinal features are present. Although the specific cause is unknown, it is likely to be the result of immune hyperreactivity to a variety of triggers in a genetically susceptible host (a polymorphism in the *ITPKC* gene, a negative regulator of T-cell activation on chromosome 19 is strongly associated with susceptibility to the disease).

There is no diagnostic test; instead, the diagnosis is made on clinical findings (Fig. 14.17). In addition to the classic features, affected children are strikingly irritable, have a high fever that is difficult to control, and may also have inflammation of their BCG vaccination site. They have high inflammatory markers (C-reactive protein, ESR, white cell count), with a platelet count that rises typically in the second week of the illness. The coronary arteries are affected in about one-third of affected children within the first 6 weeks of the illness. This can lead to aneurysms which are best visualised on echocardiography (see Case History 14.2). Subsequent narrowing of the vessels from scar formation can result in myocardial ischaemia and sudden death. Mortality is 1–2%.

Prompt treatment with intravenous immunoglobulin (IVIG) given within the first 10 days has been shown to lower the risk of coronary artery aneurysms. Aspirin is used to reduce the risk of thrombosis. It is given at a high anti-inflammatory dose until the fever subsides and inflammatory markers return to normal, and continued at a low antiplatelet dose until echocardiography at 6 weeks reveals the presence or absence of aneurysms. When the platelet count is very high, antiplatelet aggregation agents may also be used to reduce the risk of coronary thrombosis. Children with giant coronary artery aneurysms may require long-term warfarin therapy and close follow-up. Children suspected of having the disease but who do not have all the clinical features should still be considered for treatment. Sometimes, fever recurs despite treatment and these children are given a second dose of IVIG. Persistent inflammation and fever may require treatment with infliximab (a monoclonal antibody against TNF-α), steroids or ciclosporin.

Box 14.2 Causes of prolonged fever

Infective:

- Localised infection
- Bacterial infections: e.g. typhoid, *Bartonella henselae* (cat scratch disease), *Brucella*
- Deep abscesses: e.g. intra-abdominal, retroperitoneal, pelvic
- Infective endocarditis
- Tuberculosis
- Non-tuberculous mycobacterial infections: e.g. *Mycobacterium avium* complex
- Viral infections: e.g. EBV, CMV, HIV
- Parasitic infections: e.g. malaria, toxocariasis

Non-infective:

- Systemic juvenile idiopathic arthritis (SJIA)
- Systemic lupus erythematosus (SLE)
- Vasculitis (including Kawasaki disease)
- Inflammatory bowel disease
- Sarcoidosis
- Malignancy: e.g. leukaemia, lymphoma, neuroblastoma
- Macrophage activation syndromes: e.g. HLH (haemophagocytic lymphohistiocytosis)
- Drug fever
- Fabricated or induced illness.

Prolonged fever – check – is it Kawasaki disease?

Summary

Kawasaki disease

- Mainly affects infants and young children
- The diagnosis is made on clinical features – fever >5 days and four other features of non-purulent conjunctivitis, red mucous membranes, cervical lymphadenopathy, rash, red and oedematous palms and soles or peeling of fingers and toes
- 'Incomplete' (formerly called 'atypical') cases can occur, especially in infants, so a high index of suspicion should be maintained in a febrile child
- Complications – coronary artery aneurysms and sudden death
- Treatment – intravenous immunoglobulin and aspirin.

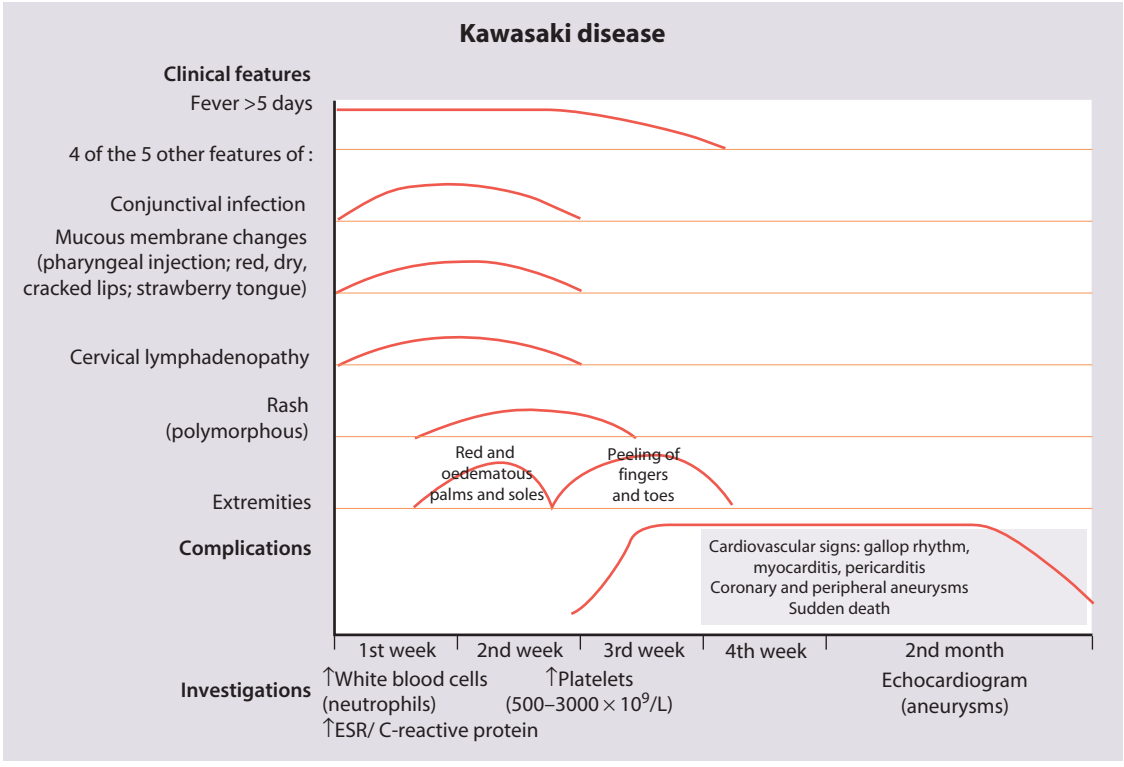


Figure 14.17 Evolution of clinical features and abnormal investigations in Kawasaki disease.

Case History

14.2 Kawasaki disease

This 2-year-old boy developed a high fever of 2 days' duration. Examination showed a miserable child with mild conjunctivitis, a rash and cervical lymphadenopathy. A viral infection was diagnosed and his mother was reassured. When he presented to hospital 3 days later, he was noted to have cracked red lips (Fig. 14.18a). He was admitted and a full septic screen, including a lumbar puncture, was performed and antibiotics started. The following day, he was still febrile and irritable; his C-reactive protein (CRP) had risen to 135, and ESR (erythrocyte sedimentation rate) to 125. Kawasaki disease was suspected and he was treated with intravenous immunoglobulin and high-dose oral aspirin. His clinical condition improved and he became afebrile the following morning. An echocardiogram at this stage showed no aneurysms of the coronary arteries, which are the most serious complication associated with delayed diagnosis and treatment. On the fifteenth day of the illness there was peeling of the fingers and toes (Fig. 14.18b).



(a) (b)

Figure 14.18 (a) Red, cracked lips and conjunctival inflammation. **(b)** Peeling of the fingers, which developed on the 15th day of the illness. (Courtesy of Professor Mike Levin.)

Tuberculosis

The decline in the incidence and mortality from tuberculosis (TB) in developed countries was hailed as an example of how public health measures and antimicrobial therapy can dramatically modify a disease. However, TB is again becoming a public health problem, partly through its increasing incidence in patients with HIV infection, and with the emergence of multidrug resistant strains. Spread of TB is usually by the respiratory route. Close proximity, infectious load and underlying immunodeficiency enhance the risk of transmission. There is an important distinction between TB *infection* (latent TB) and TB *disease*. TB infection is more likely to progress to disease in infants and young children, compared to adults. In contrast to adults, children are generally not infectious, because the disease is paucibacillary. Children usually acquire TB from an infected adult in the household.

Clinical features

These are outlined in Figure 14.19.

Diagnosis

Diagnosis of TB in children is even more difficult than in adults. The clinical features of the disease are nonspecific, such as prolonged fever, malaise, anorexia, weight loss or focal signs of infection. Sputum samples are generally unobtainable from children under about 8 years of age, unless specialist induction techniques are used. Children usually swallow sputum, so gastric washings on three consecutive mornings are required to visualise or culture acid-fast bacilli originating from the lung. To obtain these, a nasogastric tube is passed and secretions are rinsed out of the stomach with saline before food. Urine, lymph node excision, CSF and radiological examinations should also be performed where appropriate. Although it is difficult to culture TB from children, the presence of multi-drug resistant strains makes it important to try to grow the organism so that antibiotic sensitivity can be assessed.

If TB is suspected, a Mantoux test is performed – 2 units of purified protein derivative (PPD) of tuberculin (2TU SSI, 0.1 ml intradermal injection, read after 48–72 h as induration measured in mm across the forearm). Because PPD is a mixture of proteins, some of which are common to TB and BCG, the Mantoux test may be positive because of past vaccination rather than TB infection. A history of BCG immunisation therefore needs to be taken into account when interpreting the test. Induration of >10 mm is positive where no BCG has been given, 15 mm where BCG has been given. Heaf tests are no longer used for screening for TB.

A new generation of diagnostic tests is the interferon-gamma release assays (IGRA). These are blood tests that assess the response of T cells to stimulation in vitro with a small number of antigens found in TB but not in BCG. Positive results therefore indicate TB infection rather than BCG vaccination. The sensitivity and specificity of these tests in different settings is being evaluated, but its routine use in clinical practice is increasing. Neither this test nor the Mantoux test can reliably distinguish between TB infection and TB disease, so correlation with clinical signs and symptoms is required.

Co-infection with HIV makes the diagnosis even more difficult. With advanced immune suppression, both skin tests and IGRA are unreactive. Contact history, radiology and possibly tissue diagnosis become even more important. Attention has to be paid to avoiding misdiagnosing TB on chest X-ray appearances alone, as lymphoid interstitial pneumonitis can look similar and occurs in 20% of HIV-infected children. In view of the overlapping epidemiology, all individuals with TB should be tested for HIV, and vice-versa.

Treatment

Triple or quadruple therapy (rifampicin, isoniazid, pyrazinamide, ethambutol) is the recommended initial combination. This is decreased to the two drugs rifampicin and isoniazid after 2 months, by which time antibiotic sensitivities are often known. Treatment for uncomplicated pulmonary or lymph node TB is usually for 6 months; longer treatment courses are required for TB meningitis or disseminated disease. After puberty, pyridoxine is given weekly to prevent the peripheral neuropathy associated with isoniazid therapy, a complication which does not occur in young children. In tuberculous meningitis, dexamethasone is given for the first month at least, to decrease the risk of long-term sequelae.

Asymptomatic children who are Mantoux-positive and therefore latently infected should also be treated (e.g. with rifampicin and isoniazid for 3 months) as this will decrease the risk of reactivation of infection later in life.

Prevention and contact tracing

BCG immunisation has been shown to be helpful in preventing or modifying TB in the UK. However, its usefulness worldwide in preventing the disease is controversial. In the UK, BCG is recommended at birth for high-risk groups (communities with a relatively high prevalence of TB; i.e. Asian or African origin or TB in a family member in the previous 5 years or if the local area has a high prevalence rate). The UK programme of routine BCG for all tuberculin-negative children between 10 and 14 years has been discontinued. BCG should not be given to HIV-positive or other immunosuppressed children due to the potential risk of dissemination.

As most children are infected from a household contact, it is essential to screen other family members for the disease. Children who are exposed to smear-positive individuals (where organisms are visualised on sputum) should be assessed for evidence of asymptomatic infection. Mantoux-negative children over 5 years should receive BCG immunisation. Some clinicians suggest that those who are Mantoux-negative and <5 years old should receive chemoprophylaxis (e.g. rifampicin and isoniazid for 3 months). If at the end of this time they remain Mantoux-negative they should also receive BCG immunisation. Again, the aim of treatment is to decrease the risk of reactivation of TB infection later in life.

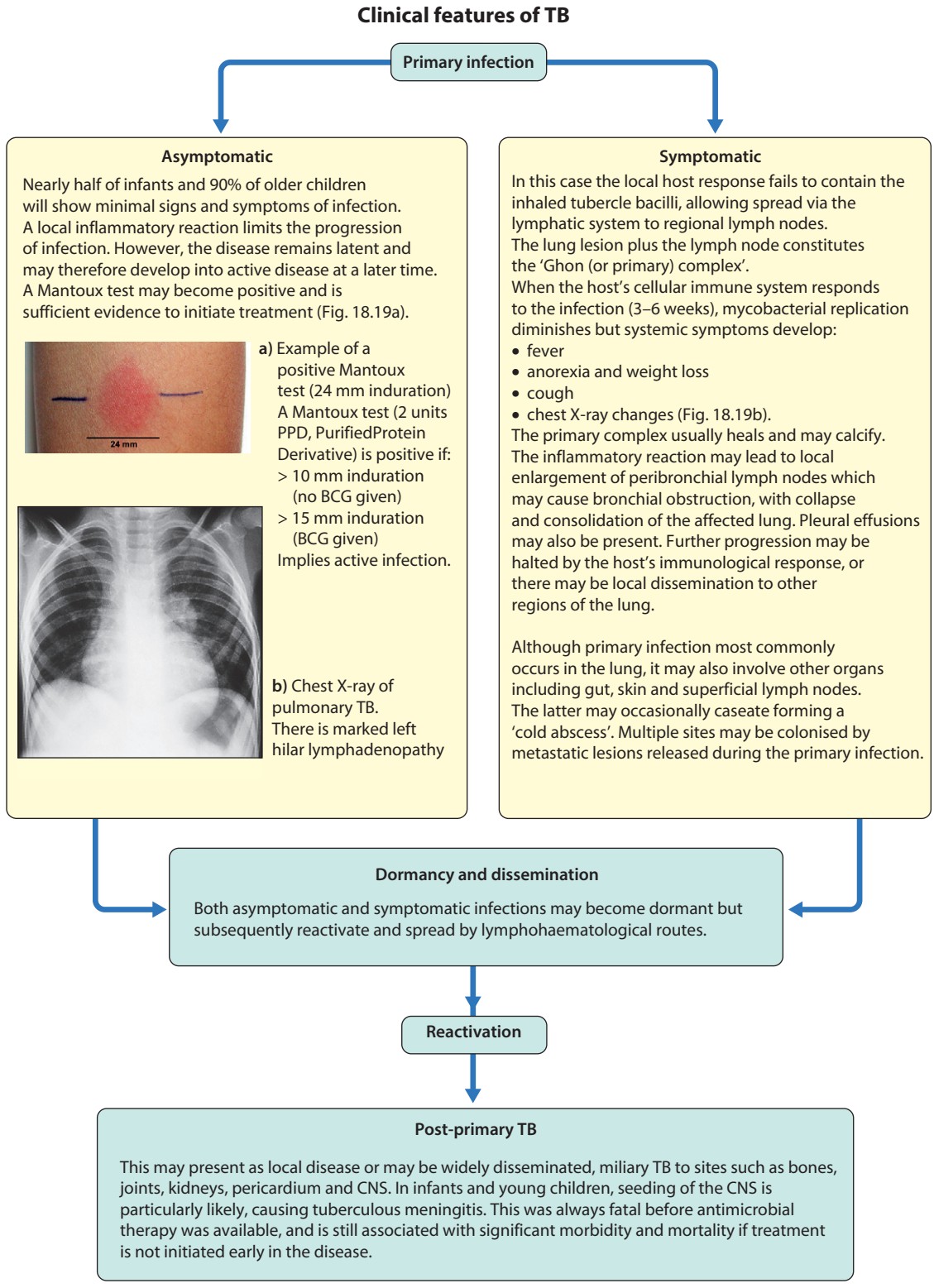


Figure 14.19 Clinical features of TB.

Non-tuberculous mycobacterial infection

There are numerous non-tuberculous mycobacteria found in the environment. Immunocompetent individuals rarely suffer from diseases caused by these organisms. They occasionally cause persistent lymphadenopathy in young children, which is usually treated surgically by complete node excision, as biopsy or partial excision may result in formation of a chronic fistula. Unlike TB, these organisms are transmitted in soil and water and therefore contact tracing is

not required following infection. Non-tuberculous mycobacteria may cause disseminated infection in immune deficient individuals. *Mycobacterium avium intracellulare* (MAI) infections are particularly common in patients with advanced HIV disease. These infections do not respond to conventional TB treatment, and require a cocktail of alternative anti-mycobacterial drugs.

Summary

Tuberculosis

- TB affects millions of children worldwide; low but increasing incidence in many developed countries
- Clinical features follow a sequence – primary infection, then dormancy, which may be followed by reactivation to post-primary TB
- Diagnosis is often difficult, so decision to treat is usually based on contact history, Mantoux test, interferon-gamma release assays (IGRA), chest X-ray and clinical features. Young children swallow their sputum, so gastric washings are required
- Adherence to drug therapy can be problematic but is essential for successful treatment
- Contact tracing is important
- TB is more difficult to diagnose and more likely to disseminate in the immunosuppressed.

Tropical infections

Although tropical infections must be considered, children who have been or are in the tropics are still susceptible to the usual range of childhood infections found. The most common or most serious imported infections are outlined in Figure 14.20.

A febrile child returning from the tropics – commonest causes are non-tropical infections, but consider malaria, typhoid fever and other tropical infections.

HIV infection

Globally, HIV infection affects over 2 million children, mostly in sub-Saharan Africa (Fig. 14.21), and there are still 380 000 children becoming infected each year (Fig. 14.22). The major route of HIV infection in children is mother-to-child transmission (MTCT), which occurs during pregnancy (intrauterine), at delivery (intrapartum) or through breast-feeding (postpartum). The virus may also be transmitted to children by infected blood products, contaminated needles or through child sexual abuse, but this is uncommon.

Diagnosis

In children over 18 months old, HIV infection is diagnosed by detecting antibodies to the virus. Children less than 18 months of age who are born to infected mothers will have transplacental maternal IgG HIV antibodies, and at this age, a positive test confirms HIV exposure but not HIV infection. The most sensitive test for HIV diagnosis before 18 months of age is HIV DNA PCR. All infants born to HIV-infected mothers should be tested for HIV infection, whether or not they are symptomatic. Two negative HIV DNA PCRs within the first 3 months of life, at least 2 weeks after completion of postnatal antiretroviral therapy, indicate the infant is not infected, although this is confirmed by the loss of transplacental maternal HIV antibodies from the infant's circulation after 18 months of age.

Clinical features

A proportion of HIV-infected infants progress rapidly to symptomatic disease and onset of AIDS in the first year of life; however, other infected children remain asymptomatic for months or years before progressing to clinical disease. Some asymptomatic children will only be identified in adolescence at routine screening following diagnosis in another family member. Clinical presentation varies with the degree of immunosuppression. Children with mild immunosuppression may have lymphadenopathy or parotitis; if moderate, they may have recurrent bacterial infections, candidiasis, chronic diarrhoea and lymphocytic interstitial pneumonitis (LIP) (Fig. 14.23). This lymphocytic infiltration of the lungs may be caused by a response to the HIV infection itself, or it may be related to EBV infection. Severe AIDS diagnoses include opportunistic infections, e.g. *Pneumocystis jiroveci* (*carinii*) pneumonia (PCP), severe failure to thrive, encephalopathy (Fig. 14.24), and malignancy, although this is rare in children. More than one clinical feature is often present. An unusual constellation of symptoms, especially if infectious, should alert one to HIV infection.

Children with persistent lymphadenopathy, hepatosplenomegaly, recurrent fever, parotid swelling, thrombocytopenia, or any suggestion of SPUR (Serious, Persistent, Unusual, Recurrent) infections should be tested for HIV.

Treatment

A decision to start antiretroviral therapy (ART) is based on a combination of clinical status, HIV viral load and CD4 count, except in infants who should all start ART shortly after diagnosis, because they have a higher risk of disease progression. As in adults, combinations of three (or four) drugs are used. Prophylaxis against *Pneumocystis jiroveci (carinii)* pneumonia (PCP), with co-trimoxazole, is prescribed for infants who are HIV-infected, and for older children with low CD4 counts.

Other aspects of management include:

- Immunisation, which is important because of the higher risk of infections, and should follow

An approach to the febrile child returning from the tropics

History

All places visited and duration of stay. Immunisation, malaria prophylaxis. History of food, drink (infected water), accommodation (exposure to vectors), contacts, swimming (infected rivers and lakes).

↓

Examination

Particular reference to: fever, jaundice, anaemia, enlarged liver or spleen.

↓

Non-tropical causes of fever

Consider non-tropical causes of fever in childhood – upper and lower respiratory tract infections, gastroenteritis, urinary tract infection, septicaemia, meningitis, osteomyelitis, hepatitis, viral infections including the childhood exanthems.

↓

Tropical infections

Malaria

40% of the world's population live in an area where the female *Anopheles* mosquito transmits malaria. Causes over 700,000 child deaths in Africa each year, predominantly from *Plasmodium falciparum* malaria. The clinical features include fever (often not cyclical), diarrhoea, vomiting, flu-like symptoms, jaundice, anaemia and thrombocytopenia.

Whilst typically the onset is 7–10 days after inoculation, infections can present many months later.

Children are particularly susceptible to severe anaemia and the gravest form of the disease, cerebral malaria.

The infection is diagnosed by examination of a thick film. The species (*falciparum, vivax, ovale* or *malariae*) is confirmed on a thin film. Repeated blood films may be necessary.

Quinine is required in most cases of *Plasmodium falciparum* seen in the UK because of the emergence of chloroquine-resistant strains worldwide.Travellers to endemic areas should always seek up-to date information on malaria prevention. Prophylaxis reduces but does not eliminate the risk of infection. Prevention of mosquito bites with repellants and bed nets is also important. In many countries there has been a marked reduction in the incidence of malaria in children from insecticide-treated bed nets, indoor residual spraying of houses with insecticides, destruction of mosquito larvae and breeding areas and prompt treatment with artemisinin-based combination therapy.

Typhoid

A child with worsening fever, headaches, cough, abdominal pain, anorexia, malaise and myalgia may be suffering from infection with *Salmonella typhi* or *paratyphi*. Gastrointestinal symptoms (diarrhoea or constipation) may not appear until the second week. Splenomegaly, bradycardia and rose-coloured spots on the trunk may be present. The serious complications of this disease include gastrointestinal perforation, myocarditis, hepatitis and nephritis.

The recent increase in multi-drug resistant strains, particularly from the Indian subcontinent, means that treatment with cotrimoxazole, chloramphenicol or ampicillin may be inadequate. A third-generation cephalosporin or azithromycin is usually effective.

Dengue fever

This viral infection is widespread in the tropics, and it is transmitted by mosquitoes. The primary infection is characterised by a fine erythematous rash, myalgia, arthralgia and high fever. After resolution of the fever, a secondary rash with desquamation may occur. Dengue haemorrhagic fever, also known as dengue shock syndrome, occurs when a previously infected child has a subsequent infection with a serologically different strain of the virus. Unfortunately, the partially effective host immune response serves to augment the severity of the infection. The child presents with severe capillary leak syndrome leading to hypotension as well as haemorrhagic manifestations. With fluid resuscitation, most children will recover fully. A patient with this condition is not infectious as direct person-to-person spread does not occur.

Gastroenteritis and dysentery

Gastroenteritis frequently accompanies foreign travel. 'Traveller's diarrhoea' is commonly caused by a change in gut flora, viruses including rotavirus and by *E. coli*. It rarely needs more than attention to rehydration. Fever accompanied by loose stools with blood or mucus suggests dysentery caused by *Shigella, Salmonella, Campylobacter* or *Entamoeba histolytica*. Blood cultures and stool cultures should be taken and appropriate antibiotics started, if indicated.

Viral haemorrhagic fevers

Causes include the Lassa, Marburg, Ebola and Crimean–Congo viruses. These rare infections are imported, although Hantavirus has recently been isolated from within the UK. These are highly contagious, often lethal, infections. If suspected, strict isolation procedures should be initiated for any symptomatic patient who has returned from an endemic area within the 21-day incubation period of these infections. Specialist advice should be sought.

Figure 14.20 An approach to the febrile child returning from the tropics.

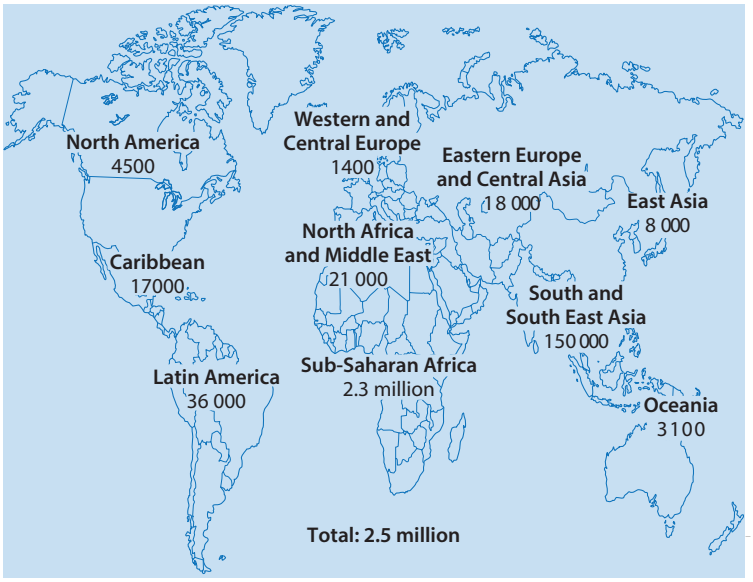


Figure 14.21 Children (<15 years) estimated to be living with HIV, 2009 (UNAIDS, WHO, 2010). Reproduced with the permission of UNICEF.

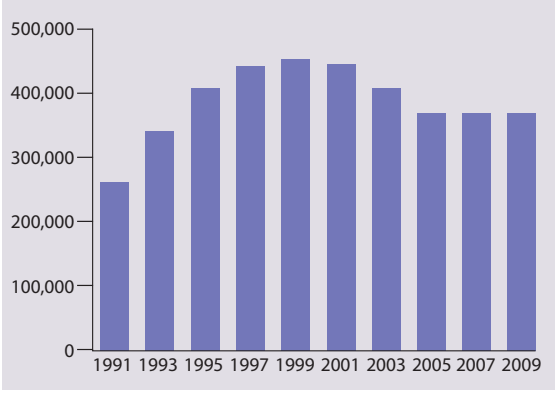


Figure 14.22 New HIV infections among children, 1991–2009, showing recent decline because HIV prevalence in mothers has stabilised and increasing programmes for preventing mother-to-child transmission. (UNAIDS, WHO, 2010; Reproduced with the permission of UNICEF.)

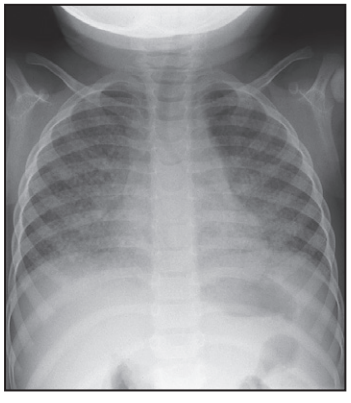


Figure 14.23 Lymphocytic interstitial pneumonitis (LIP) in a child with HIV infection. There is diffuse reticulonodular shadowing with hilar lymphadenopathy.



Figure 14.24 A CT scan in a child with HIV encephalopathy showing diffuse increase in CSF spaces from cerebral atrophy and volume loss.

the routine vaccination schedule, with the exception of BCG which should not be given as it is a live vaccine that can cause disseminated disease. Additional vaccination against influenza, hepatitis A, B and varicella zoster should be considered.

- Multidisciplinary management of children, if possible in a family clinic, where they can be seen together with other members of their family who may be HIV-infected and where the team includes an adult specialist. The team will need to address issues such as adherence to medication, disclosure of HIV diagnosis and planning for the future.
- Regular follow-up, with particular attention paid to weight, neurodevelopment and clinical signs and symptoms of disease. Effective antiretroviral therapy has transformed HIV infection into a chronic disease of childhood. Paediatric HIV clinics increasingly manage adolescents when there may be issues relating to maintaining ART adherence and address maternal issues such as safe sex practices, fertility and pregnancy.

Reduction of vertical transmission

Mothers who are most likely to transmit HIV to their infants are those with a high HIV viral load and more advanced disease. Where mothers breast-feed, 25–40% of infants become infected with HIV and it is known that avoidance of breast-feeding reduces the rate of transmission. In developed countries, perinatal transmission of HIV has been reduced to <1% by using a combination of interventions:

- Use of maternal antenatal, perinatal and postnatal antiretroviral drugs to achieve an undetectable maternal viral load at the time of delivery
- Avoidance of breast-feeding
- Active management of labour and delivery, to avoid prolonged rupture of the membranes or unnecessary instrumentation
- Pre-labour Caesarean section if the mother's viral load is detectable close to the time of delivery.

This effective combination of interventions is not available to all women globally. Avoidance of breast-feeding is not safe in many parts of the world, where use of formula-feeding increases the risk of gastroenteritis and malnutrition. It may be safer for babies in this environment to breast-feed, and antiretroviral drugs may be given to the breast-feeding baby or mother to reduce the ongoing risk of mother-to-child transmission through this route.

Antenatal antiretroviral drugs, active management of labour and delivery, and avoidance of breast-feeding can reduce vertical transmission of HIV to <1%.

Summary

HIV

- Affects >2 million children worldwide
- Treatment includes combination antiretroviral therapy and prophylaxis against *Pneumocystis jiroveci* pneumonia (PCP)
- The majority of perinatally infected children are surviving into adulthood if ART treatment is available
- Raises complex psychosocial issues for the family and caregivers, including when and what to tell the child and siblings, confidentiality and adherence support.

Lyme disease

This disease, caused by the spirochaete *Borrelia burgdorferi*, was first recognised in 1975 in a cluster of children with arthritis in Lyme, Connecticut. It occurs in the UK. *Borrelia burgdorferi* is transmitted by the hard tick, which has a range of hosts but favours deer and moose. Infections occur most commonly in the summer months in susceptible persons in rural settings.

Clinical features

Following an incubation period of 4–20 days, an erythematous macule at the site of the tick bite enlarges to cause the classical skin lesion known as *erythema migrans*, a painless red expanding lesion with a bright red outer spreading edge. During early disease, the skin lesion is often accompanied by fever, headache, malaise, myalgia, arthralgia and lymphadenopathy. Usually, these features fluctuate over several weeks and then resolve. Dissemination of infection in the early stages is rare, but may lead to cranial nerve palsies, meningitis, arthritis or carditis.

The late stage of Lyme disease occurs after weeks to months with neurological, cardiac and joint manifestations. Neurological disease includes meningoencephalitis and cranial (particularly facial nerve) and peripheral neuropathies. Cardiac disease includes myocarditis and heart block. Joint disease occurs in about 50% and varies from brief migratory arthralgia to acute asymmetric mono- and oligoarthritis of the large joints. Recurrent attacks of arthritis are common. In 10%, chronic erosive joint disease occurs months to years after the initial attack.

Diagnosis

This is based on clinical and epidemiological features and serology. Serology may be negative in early disease, so repeat titres after 2–4 weeks are advised. Isolation of the organism is difficult.

Treatment

The drug of choice for early uncomplicated cases over 12 years of age is doxycycline, and for younger children, amoxicillin. Intravenous treatment with ceftriaxone is required for carditis or neurological disease.

Immunisation

Immunisation is one of the most effective and economic public health measures to improve the health of both children and adults. The most notable success has been the worldwide eradication of smallpox achieved in 1979, but the prevalence of many other diseases has been dramatically reduced.

Differences exist in the composition and scheduling of immunisation programmes in different countries, and schedules change as new vaccines become available. The current UK schedule (Fig. 14.25) is available on the Department of Health website.

Features are:

- In the newborn – BCG is given to infants at high risk of infection
- At 2, 3 and 4 months of age – the '5 in 1' vaccine is given, against diphtheria, tetanus, pertussis, *H. influenzae* type b (Hib) and polio. The oral, live polio vaccine has been replaced by killed-vaccine given by injection, owing to the risk of vaccine-associated polio in unvaccinated family members or immune-deficient people following contact with gastrointestinal excretions of vaccine recipients
- At 2, 4 and 13 months, the pneumococcal conjugate vaccine (PCV13) is given

Immunisation schedule in the UK (2011)

	Birth	1 month	2 months	3 months	4 months	12-13 months	3 years 4 months +	12–13 years	13–18 years
BCG	BCG if at risk								
Hep B	Hep B if at risk	Hep B if at risk	Hep B if at risk			Hep B if at risk			
Diphtheria, tetanus, pertussis, polio, Haemophilus influenzae type B (DTaP/IPV/Hib)			5 in 1	5 in 1	5 in 1				
Pneumococcal vaccine (PCV)			Pneumo vaccine		Pneumo vaccine	Pneumo vaccine			
Meningococcal group C (MenC)				MenC	MenC				
Hib/MenC						Hib/Men C			
MMR						MMR	MMR		
Diphtheria, tetanus, pertussis, polio, (DTaP/IPV)							DTaP/IPV		
Human papilloma-virus (HPV) Girls only								HPV (3 doses)	
Diphtheria tetanus, polio (Td/IPV)									Td/IPV

Figure 14.25 Immunisation schedule in the UK. (Available at: www.dh.gov.uk, Accessed August 2011.)

- At 3 and 4 months, the conjugate vaccine against group C meningococcus (MenC) is given by separate injection
- At 12–13 months, a booster Hib vaccine is given, MenC and MMR (measles, mumps, rubella) is given
- At 12–13 years of age, the human papillomavirus (HPV) vaccine is given to girls. The rubella vaccine is no longer given to adolescent girls.
- BCG is no longer given to adolescents.

Rationale behind the immunisation programme

Diphtheria – infection causes local disease with membrane formation affecting the nose, pharynx or larynx or systemic disease with myocarditis and neurological manifestations. Immunisation has eradicated the disease in the UK (Fig. 14.26a).

Pertussis – clinical features described in Chapter 16. Huge decline in incidence with immunisation, but epidemics recur when immunisation rates fall (Fig. 14.26b).

Haemophilus influenzae type b – causes invasive disease in young children The number of reports of infection dropped dramatically after the introduction of Hib vaccination (Fig. 14.26c), but a gradual rise from 1988 occurred because protection was not maintained throughout childhood. This was managed with a Hib catch-up programme, and to prevent a further resurgence, a Hib booster dose has been introduced at 12 months of age.

Poliovirus infection – Although most infected children are asymptomatic or have a mild illness, some develop aseptic meningitis and <1% develop paralytic polio. Almost eradicated worldwide (Fig. 14.26d).

Meningococcal C – The marked fall in the number of reports in all age groups is shown in Figure 14.26e.

Pneumococcal vaccination – introduced into the immunisation programme in 2006. Prior to this, about 530 children under 2 years of age developed invasive pneumococcal disease in England and Wales each year. In 2010, a 13-valent conjugate vaccine (PCV, effective against 13 serotypes) was introduced, which protects against about 90% of disease-causing pneumococcal serotypes.

Human papillomavirus (*HPV*) *vaccine* – introduced in 2008. Provides protection against the two strains (HPV16 and 18) that cause 70% of cervical cancer. Three doses of vaccine are given over a 6-month period to all girls aged 12–13 years of age.

BCG immunisation – although the number of notifications of TB is rising, it remains uncommon and mainly confined to high-risk populations. BCG immunisation in the neonatal period is therefore targeted to those at increased risk. The main value of BCG is in prevention of disseminated disease (including meningitis) in younger children, hence the rationale for changing the timing of vaccination from early adolescence to the neonatal period.

Hepatitis B and *varicella vaccination* – included in the immunisation programme in the USA and many other countries. In the UK, hepatitis B immunisation is given to babies born to hepatitis B surface antigen positive (HBsAg) mothers at 0, 1, 2 and 12 months of age. Babies born to highly infectious e-antigen positive mothers (HBeAg positive) should additionally receive hepatitis B immunoglobulin at birth. Varicella vaccine is not routinely given in the UK, but may be given to the siblings of 'at-risk' children (e.g. those undergoing chemotherapy).

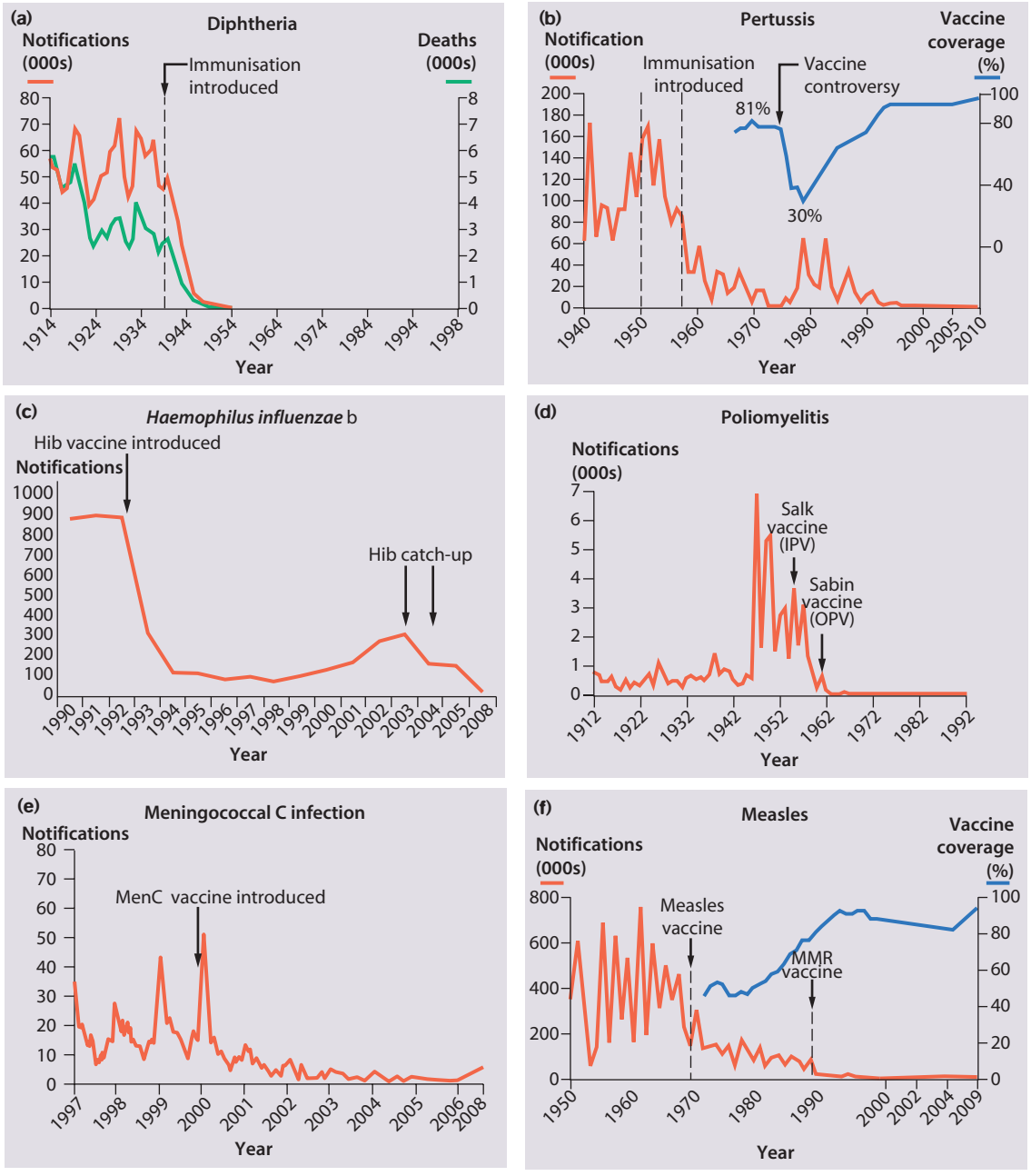


Figure 14.26 Effect of immunisation on the number of notifications in England and Wales. **(a)** Diphtheria. **(b)** Pertussis. **(c)** *Haemophilus influenzae* type b. **(d)** Poliomyelitis. **(e)** Meningococcal disease. **(f)** Measles (Health Protection Agency).

Developing countries – huge effort and funds are devoted to improving immunisation uptake and programmes. The Expanded Programme on Immunization of the WHO is tailored according to healthcare systems. The majority use measles vaccine rather than MMR, for cost–benefit reasons.

Complications and contraindications

Following vaccination, there may be swelling and discomfort at the injection site and a mild fever and malaise. Some vaccines, such as measles and rubella, may be followed by a mild form of the disease 7–10 days later. More serious reactions, including anaphylaxis, may occur but are very rare. Local guidelines about vaccination and its contraindications should be followed. Vaccination should be postponed if the child has an acute illness; however, a minor infection without fever or systemic upset is not a contraindication. Live vaccines should not be given to children with impaired immune responsiveness (except in children with HIV infection in whom MMR vaccine can be given).

The controversy regarding a possible association between MMR vaccination and autism and inflammatory bowel disease has been discredited by a large number of well-conducted studies. However, public

confidence in the immunisation programme was damaged, and uptake rates dropped (Fig. 14.26f). The MMR vaccine is only contraindicated in children with proven non-HIV-related immunodeficiency and those who are allergic to neomycin or kanamycin, which may be present in small quantities in the vaccine. Children with a history of anaphylaxis to egg (the virus is grown in fibroblast cultures generated from chick embryos) should be immunised with MMR under medical supervision. There is a 10% vaccine failure rate from primary vaccination with MMR at 12–13 months of age, but the proportion of susceptible school-age children in the UK has been reduced by the introduction of a preschool booster of MMR. Detailed information on MMR and other vaccines is available at: http://www.dh.gov.uk/en/Publichealth/Immunisation; further information about contraindications to vaccination can be found in the Department of Health Green Book, at: http://www.dh.gov.uk/en/Publichealth/Immunisation/Greenbook.

Immune deficiency

Immune deficiency may be:

- Primary (uncommon) – an intrinsic defect in the immune system
- Secondary (more common) – caused by another disease or treatment, such as an intercurrent bacterial or viral infection, malignancy, malnutrition, HIV infection, immunosuppressive therapy, splenectomy or nephrotic syndrome.

Primary immune deficiencies

Many of the primary immunodeficiencies are inherited as X-linked or autosomal recessive disorders. There may be a family history of parental consanguinity and unexplained death, particularly in boys. Immune deficiency should be considered in children who present with **S**evere, **P**rolonged, **U**nusual or **R**ecurrent (**SPUR**) infections (Box 14.3). The clinical presentation of the different primary immune deficiencies is shown in Figure 14.27.

Box 14.3 Presentation of immune deficiency

- Recurrent (proven) bacterial infections
- Severe infections (e.g. meningitis, osteomyelitis, pneumonia)
- Infections that present atypically, are unusually severe or chronic or fail regular treatment
- Infections caused by an unexpected or opportunistic pathogen or child has been immunised against
- Severe or long-lasting warts, generalised molluscum contagiosum
- Extensive candidiasis
- Complications of vaccination (disseminated BCG)
- Abscesses of internal organs; recurrent subcutaneous abscesses
- Prolonged or recurrent diarrhoea.

From: de Vries E. 2006. Patient-centred screening for primary immunodeficiency: a multi-stage diagnostic protocol designed for non-immunologists. *Clin Exp Immunol* 145:204–214.

Investigation

This is directed towards the most likely cause (Table 14.3). Investigations can quantify the essential components of the immune system and also provide a functional assessment of immunocompetence.

Table 14.3 Investigation to identify primary immune deficiency

Immune defect	Investigations
Cellular (T cells)	Full blood count (lymphocyte count)
	Lymphocyte subsets (to assess CD3+ (total T cell), CD4+ (helper T cell) and CD8+ (cytotoxic T cell) numbers)
	PHA (phytohaemagglutinin)
	Ability of T cells to proliferate in response to mitogen
Antibody (humoral; B cells)	Immunoglobulins
	IgG subclasses (in children >2 years)
	Specific antibody responses (e.g. tetanus, pneumococcal antibodies)
	Lymphocyte subsets (B cells present?)
Combined (B and T cells)	Investigations as above
	Specific genetic/molecular tests for severe combined immunodeficiency (SCID)
Neutrophils	Full blood count (neutropenia)
	Nitroblue tetrazolium (NBT) – abnormal in chronic granulomatous disease
	Tests for leucocyte adhesion deficiency – CD11b/CD18 expression
	Tests of chemotaxis (neutrophil mobility)
Complement/ mannose-binding lectin	Tests of classical and alternative complement pathways (CH50, AP50)
	Assays for individual complement proteins
	Mannose-binding lectin levels

Clinical presentation of primary immune deficiency

Defect	Presentation	Examples
T-cell defects	Severe and/or unusual viral and fungal infections and failure to thrive in first months of life, e.g. severe bronchiolitis, diarrhoea, oral thrush, *Pneumocystis jiroveci (carinii)* pneumonia (PCP).	**Severe combined immunodeficiency (SCID).** Heterogeneous group of inherited disorders of profoundly defective cellular and humoral immunity. Fatal without treatment. **HIV infection** **Wiskott-Aldrich** – triad with thrombocytopenia, and eczema (X-linked). **DiGeorge** – with maldevelopment of the 5th branchial arch causing heart defects, palatal and facial defects, absence of thymus and hypocalcaemia (deletion of section of chromosome 22). **Duncan syndrome (X-linked lymphoproliferative disease)** – inability to make a normal response to Epstein–Barr virus; either succumb to the initial infection or develop secondary lymphoma (X-linked). **Ataxia telangectasia** – defect in DNA repair, also increased risk of lymphoma. Cerebellar ataxia, developmental delay.
B-cell (antibody) defects	In first 2 years (beyond infancy because of passively acquired maternal antibody) – severe bacterial infections, especially ear, sinus, pulmonary and skin infections; recurrent diarrhoea and failure to thrive. Recurrent pneumonias can lead to bronchiectasis; recurrent ear infections to impaired hearing.	**X-linked (Bruton) agammaglobulinaemia.** Abnormal tyrosine kinase gene; essential for B-cell maturation. **Common variable immune deficiency (CVID) –** B-cell deficiency. High risk of autoimmune disorders and malignancy. Later onset than Bruton agammaglobulinaemia. **Hyper IgM syndrome** – B cells produce IgM but prevented from switching to IgG and IgA. **Selective IgA deficiency** – most common primary immune defect. Usually asymptomatic, or recurrent ear, sinus and pulmonary infections.
Neutrophil defects	Recurrent bacterial infections – abscesses (skin, lymph nodes, lung, liver, spleen, bone), poor wound healing, perianal disease and periodontal infections; invasive fungal infections, such as aspergillosis. Diarrhoea and failure to thrive. Granulomas from chronic inflammation.	**Chronic granulomatous disease** – most are X-linked recessive, some autosomal recessive. Defect in phagocytosis as fail to produce superoxide after ingestion of micro-organisms.
Leucocyte function defects	Delayed separation of umbilical cord, delayed wound healing, chronic skin ulcers and deep-seated infections.	**Leucocyte adhesion deficiency (LAD) –** deficiency of neutrophil surface adhesion molecules, CD18, CD11b, leads to inability of neutrophils to migrate to sites of infection/inflammation.
Complement defects	Recurrent bacterial infections. SLE-like illness. Recurrent meningococcal infections – with deficiency of the terminal complement components (C5b to C9).	**Early complement component deficiency** **Terminal complement component deficiency** **Mannose-binding lectin (MBL) deficiency.**

Figure 14.27 Clinical presentation of primary immune deficiency.

Management

Management options include:

- Antimicrobial prophylaxis
  - For T-cell and neutrophil defects – cotrimoxazole to prevent *Pneumocystis jiroveci* infection and itraconazole or fluconazole to prevent other fungal infections
  - For B-cell defects – antibiotic prophylaxis (e.g. azithromycin) to prevent recurrent bacterial (e.g. chest, ear, sinus) infections
- Antibiotic treatment
  - Prompt treatment of infections
  - Appropriate choice of antibiotics to cover likely organisms
  - Generally longer courses, with lower threshold for intravenous therapy
- Screening for end-organ disease
  - e.g. CT scan in children with antibody deficiency to detect bronchiectasis
- Immunoglobulin replacement therapy
  - For children with antibody deficiency
  - Can be given intravenously, which may require central venous (Portacath or Hickman) line insertion or subcutaneously
- Bone marrow transplantation
  - e.g. for SCID (severe combined immunodeficiency), chronic granulomatous disease
  - Can be matched sibling donor, matched unrelated donor or haploidentical (parental) transplant
- Gene therapy
  - For certain forms of SCID, but associated with a risk of leukaemia.

Further reading

American Academy of Pediatrics: *Report of the Committee on Infectious Diseases. 'Red Book'*, ed 27, Illinois, 2006, AAP.

Useful manual on paediatric infection and immunisation in the USA.

Feigin RD, Cherry JD: *Textbook of Pediatric Infectious Diseases*, ed 5, Philadelphia, PA, 2004, Saunders.

Large comprehensive textbook.

Websites for updates on immunisation and current information on infectious diseases (Accessed May 2011)

Meningitis Research Foundation: Available at: http://www.meningitis.org.

Useful teaching material on meningitis.

American Academy of Pediatrics: Available at: http://www.aap.org.

Centers for Disease Control, Atlanta, USA: Available at: http://www.cdc.gov.

CHIVA: Children's HIV Association of UK and Ireland. Available at: http://www.bhiva.org/chiva.

Guidelines to reduce HIV vertical transmission.

Department of Health Immunisation Information: Available at: http://www.dh.gov.uk/en/Publichealth/Immunisation.

Up-to-date information on the immunisation programme in the UK.

Health Protection Agency: Available at: http://www.hpa.org.uk.

MMR: The facts. Available at: http://www.dh.gov.uk/en/Publicationsandstatistics/Publications/PublicationsPolicyAndGuidance/DH_4002972.

Information about the MMR vaccine.

UNAIDS: Available at: http://www.unaids.org.

Worldwide information on HIV.

World Health Organization: Available at: http://www.who.int.

15

Allergy

Paediatric allergy	271	Asthma	275
Food allergy and food intolerance	273	Urticaria and angioedema	275
Eczema	275	Drug allergy	276
Allergic rhinitis and conjunctivitis (rhinoconjunctivitis)	275	Insect sting hypersensitivity	276
		Anaphylaxis	276

An abnormal immune system may result in:

- Allergic diseases
- Immune deficiencies
- Autoimmune disorders – either organ specific (e.g. type I diabetes mellitus) or systemic (e.g. systemic lupus erythematosus).

Paediatric allergy

Allergic diseases which affect children include food allergy, eczema, allergic rhinitis and conjunctivitis, asthma, urticaria, insect sting hypersensitivity and anaphylaxis. The reasons allergic diseases are important are:

- In the UK up to 40% of children have allergic rhinitis, eczema or asthma and up to 6% develop food allergy
- They are increasing in prevalence in many countries
- They are the commonest chronic diseases of childhood and the commonest cause of school absence and acute hospital admissions
- They cause significant morbidity and can be fatal, with about 20 children dying from asthma and 2 from food anaphylaxis in the UK each year.

Explanations of some of the terms used in 'allergy' are listed in Box 15.1.

Mechanisms of allergic disease

Many genes have been linked to the development of allergic disease. Polymorphisms or mutations in these genes lead to a susceptibility to allergy.

Allergic diseases occur when individuals make an abnormal immune response to harmless environmental stimuli, usually proteins. The developing immune system must be 'sensitised' to an allergen before an allergic immune response develops. However, sensitisation can be 'occult', e.g. sensitisation to egg from exposure to trace quantities of egg in maternal breast milk.

Only a few stimuli account for most allergic disease:

- Inhalant allergens, e.g. house-dust mite, plant pollens, pet dander and moulds in asthma and rhinitis and conjunctivitis
- Ingestant allergens, e.g. nuts, seeds, legumes, cow's milk, egg, seafood and fruits in acute allergic reactions or eczema
- Insect stings/bites, drugs and natural rubber latex.

Proteins with an unstable tertiary structure may be rendered non-allergenic by heat degradation or other forms of processing. For example, some children are allergic to raw apples, but can tolerate eating cooked apples.

Box 15.1 Allergy definitions

- *Hypersensitivity* – objectively reproducible symptoms or signs following exposure to a defined stimulus (e.g. food, drug, pollen) at a dose which is tolerated by normal people.
- *Allergy* – a hypersensitivity reaction initiated by specific immunological mechanisms. This can be IgE mediated (e.g. peanut allergy) or non-IgE mediated (e.g. coeliac disease).
- *Atopy* – a personal and/or familial tendency to produce IgE antibodies in response to ordinary exposures to potential allergens, usually proteins. Strongly associated with asthma, allergic rhinitis and conjunctivitis, eczema and food allergy.
- *Anaphylaxis* – a serious allergic reaction that is rapid in onset and may cause death.

Allergic immune responses are classified as IgE mediated or non-IgE mediated. IgE-mediated allergic reactions have a characteristic clinical course:

- An early phase, occurring within minutes of exposure to the allergen, caused by release of histamine and other mediators from mast cells. Causes urticaria, angioedema, sneezing and bronchospasm
- A late phase response may also occur after 4–6 hours. This causes nasal congestion in the upper airway, and cough and bronchospasm in the lower airway.

The majority of severe life-threatening allergic reactions are IgE mediated.

Non-IgE-mediated allergic immune responses have a delayed onset of symptoms and more varied clinical course.

The hygiene hypothesis

It is not clear why the prevalence of allergic diseases has increased in many countries and the speed of this change suggests an environmental cause. A consistent observation is that the risk is lower in younger children of large families and in children raised on farms. These findings have led to the hygiene hypothesis, which proposes that the increased prevalence is due to altered microbial exposure associated with modern living conditions (Fig. 15.1). Although the hypothesis remains the leading explanation for the increase in allergic disease, it is mainly supported by indirect evidence.

The allergic march

Allergic children develop individual allergic disorders at different ages:

- Eczema and food allergy usually develop in infancy; both are often present
- Allergic rhinitis and conjunctivitis and asthma occur most often in preschool and primary school years
- Rhinitis and conjunctivitis often precede the development of asthma, and in children with asthma, up to 80% have coexistent rhinitis.

The presence of eczema or food allergy in infancy is predictive of asthma and allergic rhinitis in later life. The progression is referred to as the 'allergic march'.

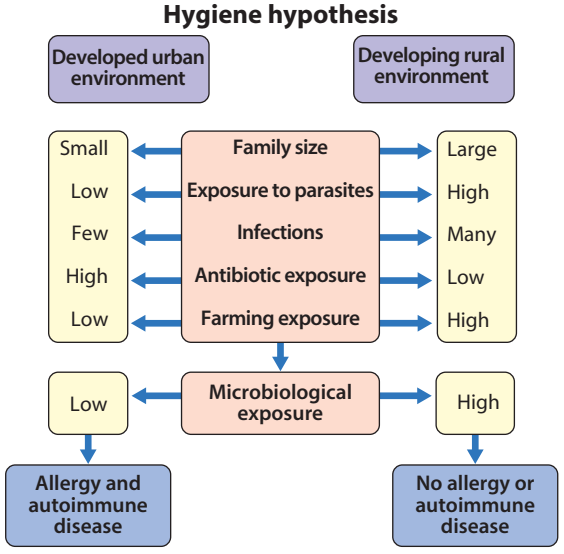


Figure 15.1 Hygiene hypothesis.

Prevention of allergic diseases

Many interventions have been tried to prevent allergic disease, or interrupt the allergic march. These include exclusive breast-feeding for at least 3–4 months (or if not possible, then use of hydrolysed formula instead of standard formula milk) to reduce the risk of eczema and cow's milk allergy and the use of probiotics for eczema in infancy. Other approaches include altering allergen exposure (avoidance of allergens in early life, or alternatively, exposure to large doses of allergens to induce immune tolerance), prebiotics (non-digestible oligosaccharides naturally present in breast milk), nutritional supplements (e.g. omega-3 fatty acids, vitamin D, antioxidants, trace elements) and medications (e.g. antihistamines, immunotherapy). However, none have been shown, long term, to prevent children from developing allergic diseases.

History and examination

The child and family may not volunteer a history of allergic disease as they have come to consider the symptoms as normal, e.g. the child who coughs most nights or has a blocked nose most of the time may not perceive this as abnormal. As allergic diseases are multisystem, in addition to the signs of individual allergic diseases, examination may reveal:

- Mouth breathing (Fig. 15.2a). Children who habitually breathe with their mouth open may have an obstructed nasal airway from rhinitis, and there may also be a history of snoring or obstructive sleep apnoea
- An allergic salute (Fig. 15.2b), from rubbing an itchy nose
- Pale and swollen inferior nasal turbinates
- Hyperinflated chest or Harrison sulci from chronic undertreated asthma
- Atopic eczema affecting the limb flexures
- Allergic conjunctivitis; may also be prominent creases (Dennie-Morgan folds) and blue-grey discoloration below the lower eyelids.

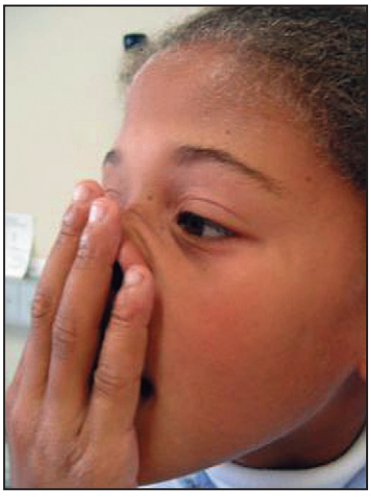
Growth needs to be checked, especially in those with food allergy, where dietary restrictions or malabsorption can lead to nutritional compromise, and in those treated with high-dose inhaled/nasal/topical corticosteroids.

Management

The individual diseases are managed by general practitioners, general paediatricians or organ-specific specialists, e.g. eczema by dermatologists, asthma by respiratory paediatricians. However, allergic diseases



(a)



(b)

Figure 15.2 Allergic facies. **(a)** There is a habitually open mouth due to mouth breathing. **(b)** An allergic salute, from rubbing an itchy nose. (Courtesy Dr George Du Toit.)

coexist and it is therefore helpful to consider allergy as a systemic disease. The role of paediatric allergists is to identify triggers to avoid, and to manage children with multisystem or severe disease.

Management of specific conditions is described below. In addition, specific allergen immunotherapy can be used for treating allergic rhinitis and conjunctivitis, insect stings, anaphylaxis and asthma. During immunotherapy, solutions of an allergen to which the patient is allergic are injected subcutaneously or administered sublingually on a regular basis for 3–5 years, with the aim of developing immune tolerance. It is highly effective in providing protection for many years. However, it must be carried out under specialist supervision due to the risk of inducing severe allergic reactions (anaphylaxis). Allergen immunotherapy is widely used in the USA and some countries in Europe. Sublingual immunotherapy appears to be safer than subcutaneous injections and is used increasingly. Immunotherapy for food allergy is under investigation but has not yet been shown to be safe for use in clinical practice.

Summary

Paediatric allergy

- Includes food allergy, eczema, allergic rhinitis and conjunctivitis, asthma, urticaria, insect sting hypersensitivity and anaphylaxis
- Occurs when a genetically susceptible person reacts abnormally to an environmental antigen
- There is an 'allergic march' of disorders
- Different allergic diseases often coexist – if a child has one, look for others.

Food allergy and food intolerance

A food allergy occurs when a pathological immune response is mounted against a specific food protein. It is usually IgE mediated, but may be non-IgE mediated. A non-immunological hypersensitivity reaction to a specific food is called food intolerance. An example of each in relation to cow's milk is shown in Figure 15.3.

Food allergy is usually primary, where children have failed to ever develop immune tolerance to the relevant food. Presentation varies with the agent and the child's age:

- In infants, the most common causes are milk, egg and peanut
- In older children, peanut, tree nut and fish and shellfish.

Food allergy can also be secondary, where children initially tolerate a food and then later become allergic to it. Secondary food allergy is usually due to cross-reactivity between proteins present in fresh fruits/vegetables/nuts and those present in pollens, e.g. a child who can eat apples may develop allergy to apples in the future if they develop allergy to birch tree pollen, because the apple and birch pollen share a very similar protein. This is termed the 'oral allergy syndrome' or 'pollen fruit syndrome'. It is very common but generally leads to milder allergic reactions than primary food allergy, often causing an itchy mouth but no systemic symptoms.

Non-IgE food allergy typically occurs hours after ingestion and usually involves the gastrointestinal tract.

Food allergy and intolerance are different from food aversion, where the person refuses the food for psychological or behavioural reasons.

Clinical features

In IgE-mediated food allergy there is a history of allergic symptoms varying from urticaria to facial swelling to anaphylaxis (Fig. 15.3), usually occurring 10–15 min after ingestion of a food. It is often the first occasion the food is knowingly ingested.

Non-IgE-mediated food allergy usually presents with diarrhoea, vomiting, abdominal pain and sometimes failure to thrive. Colic or eczema may also be present. It sometimes presents with blood in the stools in the first few weeks of life from proctitis.

Examples of food allergy and hypersensitivity to milk

Condition	Clinical manifestation	
IgE-mediated food allergy • Immediate cow's milk allergy	This 6-month-old breast-fed infant developed an allergic reaction **(a)**, with widespread urticaria immediately after the first formula feed. Skin-prick test were strongly positive to cow's milk. Widespread urticaria and lip swelling after milk ingestion are shown in **(b)** and **(c)**	**(a) Clinical features of an acute allergic reaction:** **Mild reaction** • Urticaria and itchy skin • Facial swelling **Severe reaction** • Wheeze • Stridor • Abdominal pain, vomiting, diarrhoea • Shock, collapse
		
Non-IgE-mediated cow's milk allergy	A 4-month-old infant, formula fed since birth, has loose stools and is failing to thrive. Skin prick test to cow's milk is negative. Elimination of cow's milk results in resolution of symptoms which return on trial re-introduction.	
Non-allergic food hypersensitivity • Temporary lactose intolerance	Previously well 12-month-old infant develops diarrhoea and vomiting. The vomiting settles but watery stools continue for several weeks. Stool sample – no pathogens but positive for reducing substances. Diagnosis – temporary lactose intolerance.	

Figure 15.3 Examples of food allergy and hypersensitivity to milk. **(a)** Clinical features of an acute allergic reaction. **(b,c)** Widespread urticaria and lip swelling after milk ingestion. (Courtesy Dr Pete Smith.)



Figure 15.4 Skin-prick testing for IgE-mediated allergy. A drop of the allergen is placed on the skin, the site is marked and pricked with a needle, and any weals measured. Multiple positive results are present. (Courtesy Dr Pete Smith.)

Diagnosis

The most helpful screening tests for IgE-mediated food allergy are skin-prick tests (Fig. 15.4) and measurement of specific IgE antibodies in blood (RAST test). Both tests may yield false-positive results, but the greater the response, the more likely the child is to be allergic. Negative skin test results make IgE-mediated allergy unlikely.

Non-IgE-mediated food allergies are harder to diagnose. Diagnosis relies on clinical history and examination. If indicated, endoscopy and intestinal biopsy may be obtained; the diagnosis is supported by the presence of eosinophilic infiltrates.

For both IgE-mediated and non-IgE-mediated food allergies, the gold standard investigation in cases of doubt is exclusion of the relevant food under a dietitian's supervision, followed by a double-blind placebo-controlled food challenge. This involves the child being given increasing amounts of the food or placebo, starting with a tiny quantity, until a full portion is reached. The test should be performed in hospital with full resuscitation facilities available, and close monitoring for signs of an allergic reaction.

Management

The management of a food-allergic child involves avoidance of the relevant foods. This can be very

difficult as the relevant food(s) may be present in small quantities in many foods and food labels are often unclear. Food labelling in the European Union legally requires common food allergens to be clearly disclosed. Food which is packaged or sold elsewhere may be less closely regulated. The advice of a paediatric dietitian is essential to aid patients avoid foods to which they are allergic and avoid nutritional deficiencies.

In addition, the child and family must be able to manage an allergic attack. Written self-management plans and adequate training are essential. Drug management for mild reactions (no cardiorespiratory symptoms) is with antihistamines. If the child has a severe reaction, treatment is with epinephrine (adrenaline) given intramuscularly by auto-injector (e.g. Epipen or Anapen), which the child or parent should carry with them at all times.

Food allergy to cow's milk and egg often resolves in early childhood; food allergy to nuts and seafood usually persists through to adulthood.

Summary

Food allergy

- Affects up to 6% of children
- The most common causes are milk, egg, nuts, seafood, wheat, legumes, seeds and fruits
- Diagnosis of IgE-mediated food allergy is based on a suggestive history supported by skin-prick tests or specific IgE antibodies in blood
- Supervised food challenge is sometimes necessary to clarify the diagnosis
- Those at risk of a severe reaction, e.g. with coexistent asthma, should carry an epinephrine (adrenaline) auto-injector.

Eczema

Eczema is classified as atopic (where there is evidence of IgE antibodies to common allergens) or non-atopic. Atopic eczema is classified as an allergic disease as many affected children will have a family history of allergy, at least 50% develop other allergic diseases and IgE antibodies to common allergens are present. There is a close relationship between eczema and food allergy, particularly in young infants with severe disease; up to 40% of them have an IgE-mediated food allergy, in particular egg allergy. Screening by skin prick or IgE blood testing should be considered. The condition is described in Chapter 24.

Allergic rhinitis and conjunctivitis (rhinoconjunctivitis)

This can be atopic (associated with IgE antibodies to common inhalant allergens) or non-atopic. It is an underestimated cause of childhood morbidity. The disease can be classified as intermittent or persistent and mild or severe, although in temperate climates it is often classified as seasonal (related to seasonal grass, weed or tree pollens) and perennial (related to perennial allergens such as house-dust mite and pets). It affects up to 20% of children and can severely disrupt their lives. In addition to its classic presentation of coryza and conjunctivitis, it can also present as 'cough-variant rhinitis' due to a post-nasal drip, and as a chronically blocked nose causing sleep disturbance and impaired daytime behaviour and concentration, or with predominant eye symptoms. It is associated with eczema, sinusitis and adenoidal hypertrophy and is closely associated with asthma. Treatment of allergic rhinitis may improve the control of coexistent asthma. Treatment options are listed in Box 15.2.

Box 15.2 Range of treatment for allergic rhinoconjunctivitis

- Second-generation non-sedating antihistamines (used topically or systemically)
- Topical corticosteroid nasal or eye preparations (the latter under specialist ophthalmology supervision)
- Cromoglycate eye drops
- Leukotriene receptor antagonists, e.g. montelukast
- Nasal decongestants (use for no more than 7–10 days due to risk of rebound effect)
- Allergen immunotherapy – sublingual or subcutaneous (limited by anaphylaxis risk)
- Systemic corticosteroids should not be used due to the risk of adverse effects.

Asthma

Allergy is an important component of asthma. Affected children often have IgE antibodies to aeroallergens (house-dust mite; tree, grass and weed pollens; moulds; animal danders). Allergen avoidance is difficult to achieve. Management of asthma is described in Chapter 16 on Respiratory disorders.

Urticaria and angioedema

Acute urticaria usually results from exposure to an allergen or a viral infection, which triggers an urticarial skin reaction. It may also involve deeper tissues to produce swelling of the lips and soft tissues around the eyes (angioedema), and even anaphylaxis.

Chronic urticaria (persisting >6 weeks) is usually non-allergic in origin. It results from a local increase in the permeability of capillaries and venules. These changes are dependent on activation of skin mast cells, which contain a range of mediators including histamine. A classification of urticaria is shown in Box 15.3. Treatment is with second-generation non-sedating antihistamines.

Box 15.3 Classification of urticaria/angioedema

- Acute – resolve within 6 weeks; allergy such as food or drug reactions, or infection are common triggers
- Chronic idiopathic – intermittent for at least 6 weeks
- Physical urticarias
  - Cold, delayed pressure, heat contact, solar, vibratory urticaria
- Other causes
  - Water (aquagenic), sweating (cholinergic), exercise-induced
  - Aspirin and other non-steroidal anti-inflammatory agents
  - C1-esterase inhibitor deficiency (angioedema, but no urticaria or pruritus).

Drug allergy

Drug allergies do occur in children, especially to antibiotics, but only a minority who are labelled 'drug allergic' are truly allergic. This is usually because viral illnesses, for which children are often prescribed antibiotics, themselves cause skin rashes. A detailed history is required of the nature and timing of the rash in relation to taking the antibiotics.

Allergy skin and blood tests can be used to support a diagnosis of drug allergy, but a drug challenge may be the only way to conclusively confirm or refute the diagnosis. This is contraindicated after a severe allergic reaction and an alternative drug should be sought.

Insect sting hypersensitivity

This arises mainly from bee and wasp stings, but also from ant species in the USA, Asia and Australia. The severity of the allergic reaction may be:

- mild – local swelling
- moderate – generalised urticaria
- severe – systemic symptoms with wheeze or shock.

Children with a previous mild or moderate reaction are unlikely to develop a severe reaction in the future and the families can be reassured. Those who had a severe reaction should carry an epinephrine (adrenaline) auto-injector, and allergen immunotherapy should be considered.

Summary

Insect sting hypersensitivity

- Mainly to bee and wasp stings
- Following a severe reaction, an epinephrine (adrenaline) auto-injector should be carried
- Immunotherapy is highly effective in children who have had a severe reaction.

Anaphylaxis

This serious and potentially life-threatening allergic reaction is described in Chapter 6 on Paediatric emergencies.

Further reading

Kay AB: Advances in immunology: allergy and allergic disease (first of two parts). *New England Journal of Medicine* 344(1):30–37, 2001.

Lack G: Clinical practice. Food allergy. *New England Journal of Medicine* 359(12):1252–1260, 2008.

Strachan DP: Family size, infection and atopy: the first decade of the 'hygiene hypothesis'. *Thorax* 55(Suppl 1):S2–S10, 2000.

Websites (Accessed April 2011)

Food Allergy and Anaphylaxis Network: Available at http://www.foodallergy.org

A non-profit organisation for people with food allergy.

The UCB Institute of Allergy: Available at http://www.theucbinstituteofallergy.com/healthprofessionals/knowbetterallergy/whatisallergy/index.asp

An industry-supported non-profit organisation for allergy education.

The Anaphylaxis Campaign: Available at: http://www.anaphylaxis.org.uk.

A charity for people at risk from anaphylaxis.

16

Respiratory disorders

Respiratory infections	277	Cystic fibrosis	294
Asthma	285	Sleep-related breathing disorders	298
Recurrent or persistent cough	293	Tracheostomy	299
Chronic lung infection	294	Long-term ventilation	299

Notable features of respiratory disorders are:

- Worldwide, respiratory disorders are the most common cause of death in children
- In the UK, respiratory disorders account for 50% of consultations with general practitioners for acute illness in young children and a third of consultations in older children
- Respiratory disorders are responsible for about 25% of acute paediatric admissions to hospital, some of which are life-threatening
- Asthma is the most common chronic illness of childhood in the UK
- Cystic fibrosis is the most common inherited life-limiting disorder in Caucasians.

Respiratory infections

These are the most frequent infections of childhood. The preschool child has, on average, 6–8 respiratory infections a year. Most are mild self-limiting illnesses of the upper respiratory tract (ear, nose, throat) but some, such as bronchiolitis or pneumonia, are potentially life-threatening.

Pathogens

Viruses cause 80–90% of childhood respiratory infections. The most important are the respiratory syncytial virus (RSV), rhinoviruses, parainfluenza, influenza, metapneumovirus and adenoviruses. An individual virus can cause several different patterns of illness, e.g. RSV can cause bronchiolitis, croup, pneumonia or a common cold.

The important bacterial pathogens of the respiratory tract are *Streptococcus pneumoniae* (pneumococcus) and other streptococci, *Haemophilus influenzae*, *Moraxella catarrhalis*, *Bordetella pertussis*, which causes whooping cough, and *Mycoplasma pneumoniae*. Dual infections, with two viruses or with a viral and bacterial pathogen, may occur. *Mycobacterium tuberculosis* remains an important pathogen globally. Some pathogens cause predictable epidemics, such as RSV bronchiolitis every winter.

Host and environmental factors

An increased risk of respiratory infection is associated with a range of factors relating to the environment and host:

- Parental smoking, especially maternal
- Poor socioeconomic status – large family size, overcrowded, damp housing
- Poor nutrition
- Underlying lung disease – such as bronchopulmonary dysplasia in infants who were born preterm, cystic fibrosis or asthma
- Male gender
- Haemodynamically significant congenital heart disease
- Immunodeficiency (either primary, see Ch. 14, or secondary, e.g. from HIV infection or chemotherapy).

The child's age influences the prevalence and severity of infections (Fig. 16.1). It is in infancy that serious respiratory illness requiring hospital admission is most common and the risk of death is greatest. There is an increased frequency of infections when the child or older siblings start nursery or school. Repeated upper respiratory tract infection is common and rarely indicates underlying disease.

Classification of respiratory infections

Respiratory infections are classified according to the level of the respiratory tract most involved:

- Upper respiratory tract infection
- Laryngeal/tracheal infection
- Bronchitis
- Bronchiolitis
- Pneumonia.

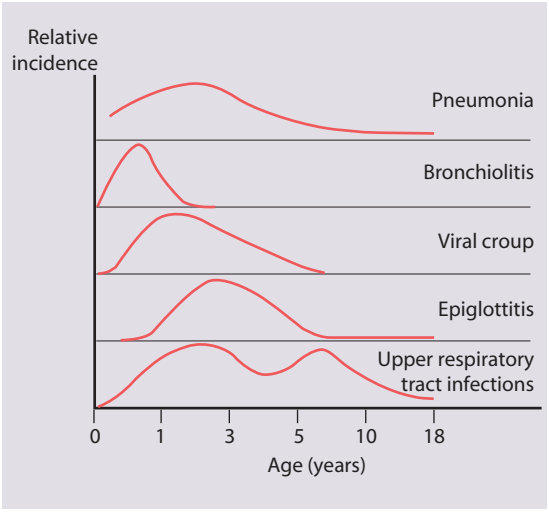


Figure 16.1 Age distribution of acute respiratory infections in children.

Upper respiratory tract infection (URTI)

Approximately 80% of all respiratory infections involve only the nose, throat, ears or sinuses. The term URTI embraces a number of different conditions:

- Common cold (coryza)
- Sore throat (pharyngitis, including tonsillitis)
- Acute otitis media
- Sinusitis (relatively uncommon).

The commonest presentation is a child with a combination of nasal discharge and blockage, fever, painful throat and earache. Cough may be troublesome. URTIs may cause:

- Difficulty in feeding in infants as their noses are blocked and this obstructs breathing
- Febrile convulsions
- Acute exacerbations of asthma.

In infants, hospital admission may be required to exclude a more serious infection, if feeding is inadequate, or for parental reassurance.

The common cold (coryza)

This is the commonest infection of childhood. Classical features include a clear or mucopurulent nasal discharge and nasal blockage. The commonest pathogens are viruses – rhinoviruses (of which there are over 100 different serotypes), coronaviruses and RSV. Health education to advise parents that colds are self-limiting and have no specific curative treatment may reduce anxiety and save unnecessary visits to doctors. Fever and pain are best treated with paracetamol or ibuprofen. Antibiotics are of no benefit as the common cold is viral in origin and secondary bacterial infection is very uncommon.

Sore throat (pharyngitis)

The pharynx and soft palate are inflamed and local lymph nodes are enlarged and tender. Sore throats are usually due to viral infection with respiratory viruses (mostly adenoviruses, enteroviruses and rhinoviruses). In the older child, group A β-haemolytic streptococcus is a common pathogen.

Tonsillitis

Tonsillitis is a form of pharyngitis where there is intense inflammation of the tonsils, often with a purulent exudate. Common pathogens are group A β-haemolytic streptococci and the Epstein–Barr virus (infectious mononucleosis). Group A β-haemolytic streptococcus can be cultured from many tonsils; however, it is uncertain why it causes recurrent tonsillitis in some children but not in others.

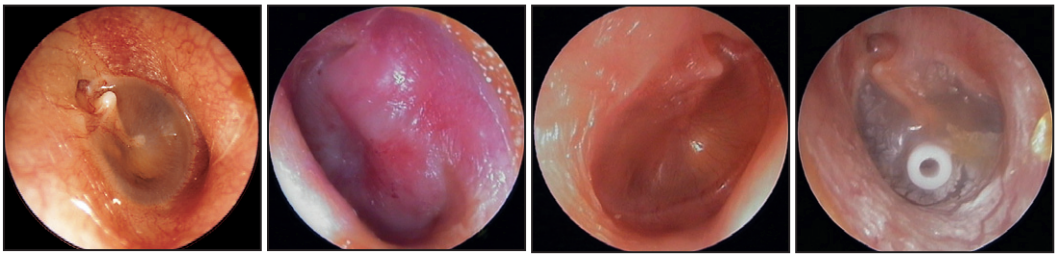
Although the surface exudates seen in infectious mononucleosis are reported to be more membranous in appearance compared to bacterial tonsillitis, in reality it is not possible to distinguish clinically between viral and bacterial causes. Marked constitutional disturbance, such as headache, apathy and abdominal pain, white tonsillar exudate and cervical lymphadenopathy, is more common with bacterial infection.

Antibiotics (often penicillin, or erythromycin if there is penicillin allergy) are often prescribed for severe pharyngitis and tonsillitis even though only a third are caused by bacteria. They may hasten recovery from streptococcal infection. In order to eradicate the organism to prevent rheumatic fever, 10 days of treatment is required, but this is not indicated in the UK where rheumatic fever is now exceedingly rare. In severe cases, children may require hospital admission for intravenous fluid administration and analgesia if they are unable to swallow solids or liquids. Amoxicillin is best avoided as it may cause a widespread maculopapular rash if the tonsillitis is due to infectious mononucleosis.

It is not possible to distinguish clinically between viral and bacterial tonsillitis.

Acute infection of the middle ear (acute otitis media)

Most children will have at least one episode of acute otitis media (OM). This is most common at 6–12 months of age. Up to 20% will have three or more episodes. Infants and young children are prone to acute otitis media because their Eustachian tubes are short, horizontal and function poorly. There is pain in the ear and fever. Every child with a fever must have their tympanic membranes examined (Fig. 16.2a–d). In acute otitis media, the tympanic membrane is seen to be bright red and bulging with loss of the normal light reflection (Fig. 16.2b). Occasionally, there is acute perforation of the eardrum with pus visible in the external canal. Pathogens include viruses, especially RSV and rhinovirus, and bacteria including pneumococcus, nontypeable *H. influenzae* and *Moraxella catarrhalis*. Serious complications are mastoiditis and meningitis, but are now uncommon. Pain should be treated with an analgesic such as paracetamol or ibuprofen. Regular analgesia is more effective than intermittent (as required) and may be needed for up to a week until the acute inflammation has resolved. Most cases of acute otitis media resolve spontaneously. Antibiotics marginally shorten the duration of pain but have not been



(a) (b) (c) (d)

Figure 16.2 Appearance of the eardrum. **(a)** Normal. **(b)** Acute otitis media. **(c)** Otitis media with effusion. **(d)** Grommet. (Courtesy of Mr N Shah & Mr N Tolley.)

shown to reduce the risk of hearing loss (see Ch. 5). It is often useful to give the parents a prescription, but ask them to use it only if the child remains unwell after 2–3 days. Amoxicillin is widely used. Neither decongestants nor antihistamines are beneficial.

Recurrent ear infections can lead to otitis media with effusion (OME or glue ear or serous otitis media). Children are asymptomatic apart from possible decreased hearing. The eardrum is seen to be dull and retracted, often with a fluid level visible (Fig. 16.2c). Confirmation of otitis media with effusion can be gained by a flat trace on tympanometry, in conjunction with evidence of a conductive loss on pure tone audiometry (possible if >4 years old), or reduced hearing on a distraction hearing test in younger children. Otitis media with effusion is very common between the ages of 2 and 7 years, with peak incidence between 2.5 and 5 years. This condition usually resolves spontaneously. Cochrane reviews have shown no evidence of long-term benefit from the use of antibiotics, steroids or decongestants. Otitis media with effusion is the most common cause of conductive hearing loss in children and can interfere with normal speech development and result in learning difficulties in school. In such children insertion of ventilation tubes (grommets, Fig. 16.2d) can be beneficial, but there is evidence, again from Cochrane reviews, that adenoidectomy can offer more long-term benefit. It is believed that the adenoids can harbour organisms within biofilms that contribute to infection spreading up the Eustachian tubes. In addition, grossly hypertrophied adenoids may obstruct and affect the function of the Eustachian tubes, leading to poor ventilation of the middle ear and subsequent recurrent infections. In practice, children with recurrent URTIs and chronic glue ear that do not resolve with conservative measures undergo grommet insertion. If these problems recur after grommet extrusion, reinsertion of grommets with adjuvant adenoidectomy is usually advocated.

Summary

Acute otitis media

- Can only be diagnosed by examining the tympanic membrane
- Antibiotics marginally shorten the duration of pain but do not reduce hearing loss
- If recurrent, may result in otitis media with effusion, which may cause speech and learning difficulties from hearing loss.

Sinusitis

Infection of the paranasal sinuses may occur with viral URTIs. Occasionally there is secondary bacterial infection, with pain, swelling and tenderness over the cheek from infection of the maxillary sinus. As the frontal sinuses do not develop until late childhood, frontal sinusitis is uncommon in the first decade of life. Antibiotics and analgesia are used for acute sinusitis in addition to topical decongestants. There is some recent evidence that the concurrent use of intranasal corticosteroids or antihistamines together with antibiotics hasten recovery.

Tonsillectomy and adenoidectomy

Children with recurrent tonsillitis are often referred for removal of their tonsils, one of the commonest operations performed in children. Many children have large tonsils but this in itself is not an indication for tonsillectomy, as they shrink spontaneously in late childhood.

The indications for tonsillectomy are controversial, and must be balanced against the risks of surgery, but include:

- Recurrent severe tonsillitis (as opposed to recurrent URTIs) – tonsillectomy reduces the number of episodes of tonsillitis by a third, e.g. from three to two per year, but is unlikely to benefit mild symptoms.
- A peritonsillar abscess (quinsy)
- Obstructive sleep apnoea (the adenoids will also normally be removed).

Like the tonsils, adenoids increase in size until about the age of 8 years and then gradually regress. In young children, the adenoids grow proportionately faster than the airway, so that their effect of narrowing the airway lumen is greatest between 2 and 8 years of age. They may narrow the posterior nasal space sufficiently to justify adenoidectomy. Indications for the removal of both the tonsils and adenoids are controversial but include:

- Recurrent otitis media with effusion with hearing loss, where it gives a significant long-term additional benefit, especially if reinsertion of grommets is considered
- Obstructive sleep apnoea (an absolute indication).

Box 16.1 Differential diagnosis of acute upper airways obstruction

Common causes

- Viral laryngotracheobronchitis ('croup' – very common)

Rare causes

- Epiglottitis
- Bacterial tracheitis
- Inhalation of smoke and hot air in fires
- Trauma to the throat
- Retropharyngeal abscess
- Laryngeal foreign body
- Allergic laryngeal angioedema (seen in anaphylaxis and recurrent croup)
- Hypocalcaemia due to poor vitamin D intake
- Infectious mononucleosis causing severe lymph node swelling
- Measles
- Diphtheria

Laryngeal and tracheal infections

The mucosal inflammation and swelling produced by laryngeal and tracheal infections can rapidly cause life-threatening obstruction of the airway in young children. Several conditions can cause acute upper airways obstruction (Box 16.1). They are characterised by:

- Stridor, a rasping sound heard predominantly on inspiration
- Hoarseness due to inflammation of the vocal cords
- A barking cough like a sea lion
- A variable degree of dyspnoea.

The severity of upper airways obstruction is best assessed clinically by the degree of chest retraction (none, only on crying, at rest) and degree of stridor (none, only on crying, at rest or biphasic) (Fig. 16.3).

Severe obstruction leads to increasing respiratory rate, heart rate and agitation. Central cyanosis or drowsiness indicates severe hypoxaemia and the need for urgent intervention – the most reliable objective measure of hypoxaemia is by measuring the oxygen saturation by pulse oximetry.

Basic management of acute upper airways obstruction

- **Do not examine the throat!**
- **Reduce anxiety by being calm, confident and well organised.**
- **Observe carefully for signs of hypoxia or deterioration.**
- **If severe, administer nebulised epinephrine (adrenaline) and contact an anaesthetist.**
- **If respiratory failure develops from increasing airways obstruction, exhaustion or secretions blocking the airway, urgent tracheal intubation is required.**



Figure 16.3 The degree of subcostal, intercostal and sternal recession is a more useful indicator of severity of upper airways obstruction than the respiratory rate.

Total obstruction of the upper airway may be precipitated by examination of the throat using a spatula. One must avoid looking at the throat of a child with upper airways obstruction unless full resuscitation equipment and personnel are at hand.

Croup

With laryngotracheobronchitis, usually called croup, there is mucosal inflammation and increased secretions affecting the airway, but it is the oedema of the subglottic area that is potentially dangerous in young children because it may result in critical narrowing of the trachea. Viral croup accounts for over 95% of laryngotracheal infections. Parainfluenza viruses are the commonest cause, but other viruses, such as human metapneumovirus, RSV and influenza, can produce a similar clinical picture. Croup occurs from 6 months to 6 years of age but the peak incidence is in the second year of life. It is commonest in the autumn. The typical features are a barking cough, harsh stridor and hoarseness, usually preceded by fever and coryza. The symptoms often start, and are worse, at night.

When the upper airway obstruction is mild, the stridor and chest recession disappear when the child is at rest. The child can usually be managed at home. The parents need to observe the child closely for the signs of increasing severity. The decision to manage the child at home or in hospital is influenced not only by the severity of the illness but also by the time of day, ease of access to hospital and the child's age (with a low threshold for admission for those <12 months old, due to their narrow airway caliber), and parental understanding and confidence about the disorder.

Inhalation of warm moist air is widely used but is of unproven benefit. Oral dexamethasone, oral prednisolone and nebulised steroids (budesonide) reduce the severity and duration of croup, and the need for hospitalisation.

In severe upper airways obstruction, nebulised epinephrine (adrenaline) with oxygen by facemask provides transient improvement. Close monitoring,

along with the advice of an anaesthetist or intensivist, is imperative due to the risk of rebound symptoms once the effects of the epinephrine (adrenaline) diminish after about 2 h. Only a few children with croup require tracheal intubation since the introduction of steroid therapy. Some children have a pattern of recurrent croup, which may be related to atopy.

Bacterial tracheitis (pseudomembranous croup)

This rare but dangerous condition is similar to severe viral croup except that the child has a high fever, appears toxic and has rapidly progressive airways obstruction with copious thick airway secretions. It is caused by infection with *Staphylococcus aureus*. Treatment is by intravenous antibiotics and intubation and ventilation if required.

Acute epiglottitis

Acute epiglottitis is a life-threatening emergency due to the high risk of respiratory obstruction. It is caused by *H. influenzae* type b. In the UK and many other countries, the introduction of universal Hib immunisation in infancy has led to a >99% reduction in the incidence of epiglottitis and other invasive *H. influenzae* type b infections.

There is intense swelling of the epiglottis and surrounding tissues associated with septicaemia. Epiglottitis is most common in children aged 1–6 years but affects all age groups. It is important to distinguish clinically between epiglottitis and croup (Table 16.1), as they require quite different treatment.

The onset of epiglottitis is often very acute (see Case History 16.1), with:

- high fever in an ill, toxic-looking child
- an intensely painful throat that prevents the child from speaking or swallowing; saliva drools down the chin
- soft inspiratory stridor and rapidly increasing respiratory difficulty over hours
- the child sitting immobile, upright, with an open mouth to optimise the airway.

Table 16.1 Clinical features of croup (viral laryngotracheitis) and epiglottitis

	Croup	Epiglottitis
Onset	Over days	Over hours
Preceding coryza	Yes	No
Cough	Severe, barking	Absent or slight
Able to drink	Yes	No
Drooling saliva	No	Yes
Appearance	Unwell	Toxic, very ill
Fever	<38.5°C	>38.5°C
Stridor	Harsh, rasping	Soft, whispering
Voice, cry	Hoarse	Muffled, reluctant to speak

Case History

16.1 Acute epiglottitis

This 5-year-old girl developed a severe sore throat, drooling of saliva, a high fever and increasing difficulty breathing over 8 h (Fig. 16.4a) Epiglottitis was diagnosed and her airway was guaranteed with a nasotracheal tube. Antibiotics were started immediately (Fig. 16.4b,c). She made a full recovery.



(a) (b) (c)

Figure 16.4 Acute epiglottitis. **(a)** At presentation. **(b)** At 16 h, with nasotracheal and nasogastric tubes and an indwelling cannula for intravenous antibiotics. **(c)** At 36 h, following removal of the nasotracheal and nasogastric tubes.

In contrast to viral croup, cough is minimal or absent. Attempts to lie the child down or examine the throat with a spatula or perform a lateral neck X-ray must not be undertaken as they can precipitate total airway obstruction and death.

If the diagnosis of epiglottitis is suspected, urgent hospital admission and treatment are required. A senior anaesthetist, paediatrician and ENT surgeon should be summoned and treatment initiated without delay. The child should be transferred directly to the intensive care unit or an anaesthetic room, and must be accompanied by senior medical staff in case respiratory obstruction occurs. The child should be intubated under controlled conditions with a general anaesthetic. Rarely, this is impossible and urgent tracheostomy is life-saving. Only after the airway is secured should blood be taken for culture and intravenous antibiotics such as cefuroxime started. The tracheal tube can usually be removed after 24 h and antibiotics given for 3–5 days. With appropriate treatment, most children recover completely within 2–3 days. As with other serious *H. influenzae* infections, prophylaxis with rifampicin is offered to close household contacts.

Minutes count in acute epiglottitis.

Bronchitis

There is controversy about the term bronchitis in childhood. While some inflammation of the bronchi producing a mixture of wheeze and coarse crackles is often a feature of respiratory infections, bronchitis in children is very different from the chronic bronchitis of adults. In acute bronchitis in children, cough and fever are the main symptoms. The cough may persist for about 2 weeks, or longer with pertussis or *Mycoplasma* infections. There is no evidence that antibiotics, cough suppressants or expectorants speed recovery.

Whooping cough (pertussis)

This is a highly contagious respiratory infection caused by *Bordetella pertussis*. It is endemic, with epidemics every 3–4 years. After a week of coryza (catarrhal phase), the child develops a characteristic paroxysmal or spasmodic cough followed by a characteristic inspiratory whoop (paroxysmal phase). The spasms of cough are often worse at night and may culminate in vomiting. During a paroxysm, the child goes red or blue in the face, and mucus flows from the nose and mouth. The whoop may be absent in infants, but apnoea is a feature at this age. Epistaxis and subconjunctival haemorrhages can occur after vigorous coughing. The paroxysmal phase lasts 3–6 weeks. The symptoms gradually decrease (convalescent phase) but may persist for many months. Complications of pertussis, such as pneumonia, convulsions and bronchiectasis, are uncommon, but there is still a significant mortality, particularly in infants. Infants who have not yet completed their primary vaccination at 4 months are particularly susceptible. Infants and young children suffering severe spasms of cough or cyanotic attacks should be admitted to hospital and isolated from other children.

The organism can be identified early in the disease from culture of a per-nasal swab, although PCR is more sensitive. Characteristically, there is a marked lymphocytosis ($>15 \times 10^9$/L) on a blood count. Although erythromycin eradicates the organism, it decreases symptoms only if started during the catarrhal phase. Siblings, parents and school contacts may develop a similar cough, and close contacts should receive erythromycin prophylaxis, and unvaccinated infant contacts should be vaccinated. Immunisation reduces the risk of developing pertussis and the severity of disease in those affected, but does not guarantee protection. The level of protection declines steadily during childhood.

Summary

Pertussis

- Caused by *Bordetella pertussis*
- Paroxysmal cough followed by inspiratory whoop and vomiting; in infants, apnoea rather than whoop, which is potentially dangerous
- Diagnosis: culture of organism on per-nasal swab, marked lymphocytosis on blood film.

Bronchiolitis

Bronchiolitis is the commonest serious respiratory infection of infancy: 2–3% of all infants are admitted to hospital with the disease each year during annual winter epidemics; 90% are aged 1–9 months (bronchiolitis is rare after 1 year of age). Respiratory syncytial virus (RSV) is the pathogen in 80% of cases. The remainder are accounted for by human metapneumovirus, parainfluenza virus, rhinovirus, adenovirus, influenza virus, and *Mycoplasma pneumoniae*. Dual infection with RSV and human metapneumovirus is associated with severe bronchiolitis.

Clinical features

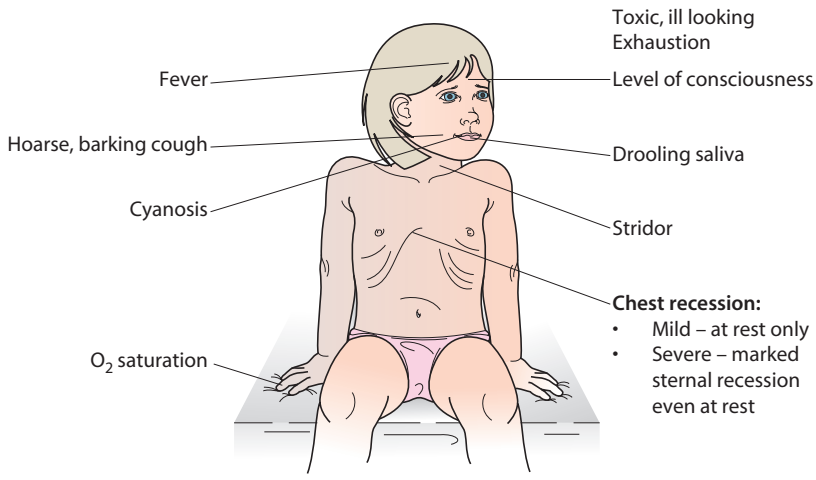
Coryzal symptoms precede a dry cough and increasing breathlessness. Feeding difficulty associated with increasing dyspnoea is often the reason for admission to hospital. Recurrent apnoea is a serious complication, especially in young infants. Infants born prematurely who develop bronchopulmonary dysplasia or with other underlying lung disease, such as cystic fibrosis or have congenital heart disease, are most at risk from severe bronchiolitis. The characteristic findings on examination (Fig. 16.5) are:

- Sharp, dry cough
- Tachypnoea
- Subcostal and intercostal recession
- Hyperinflation of the chest:
  - Prominent sternum
  - Liver displaced downwards
- Fine end-inspiratory crackles
- High-pitched wheezes – expiratory > inspiratory
- Tachycardia
- Cyanosis or pallor.

Summary

The child with stridor

Clinical features to assess



Clinical conditions

Croup
- Mostly viral
- 6 months to 6 years of age
- Harsh, loud stridor
- Coryza and mild fever, hoarse voice

Epiglottitis:
- Caused by *H. influenzae* type b, rare since Hib immunisation
- Mostly aged 1–6 years
- Acute, life-threatening illness
- High fever, ill, toxic-looking
- Painful throat, unable to swallow saliva, which drools down the chin

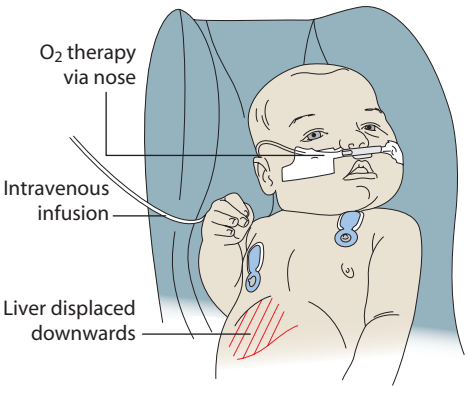
Bacterial tracheitis:
- High fever, toxic
- Loud, harsh stridor

Inhaled foreign body
- Choking on peanut or toy in mouth
- Sudden onset of cough or respiratory distress

Laryngomalacia or congenital airway abnormality:
- Recurrent or continuous stridor since birth

Other rare causes:
- See Box 16.1

Bronchiolitis



Apnoea in infants <4 months

Sharp, dry cough

Cyanosis or pallor

Hyperinflation of the chest:
• sternum prominent
• liver displaced downwards

Subcostal and intercostal recession

Auscultation:
• fine end-inspiratory crackles
• prolonged expiration

Figure 16.5 Clinical features of severe bronchiolitis in an infant.

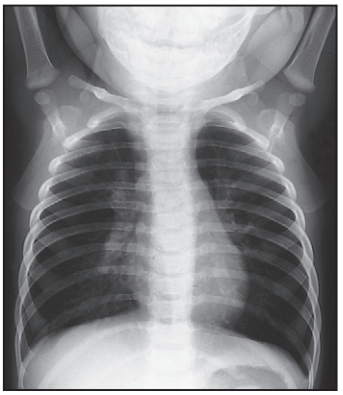


Figure 16.6 In acute bronchiolitis, the chest X-ray shows hyperinflation of the lungs with flattening of the diaphragm, horizontal ribs and increased hilar bronchial markings. However, chest X-ray is rarely helpful in bronchiolitis.

Investigations

Respiratory viruses are now usually identified by PCR analysis of nasopharyngeal secretions. A chest X-ray is unnecessary in straightforward cases, but if performed, typically shows hyperinflation of the lungs due to small airways obstruction, air trapping (Fig. 16.6) and often focal atelectasis. Pulse oximetry is used to measure and monitor arterial oxygen saturation continuously. Blood gas analysis, usually a capillary sample, is only performed in severe disease to identify hypercarbia when additional ventilatory support is considered.

Management

This is supportive. Humidified oxygen is delivered via nasal cannulae; the concentration required is determined by pulse oximetry. The infant is monitored for apnoea. Mist, antibiotics, steroids and nebulised bronchodilators, such as salbutamol or ipratropium, have not been shown to reduce the severity or duration of the illness. Fluids may need to be given by nasogastric tube or intravenously. Assisted ventilation in the form of nasal or facemask CPAP or full ventilation is required in a small percentage of infants admitted to hospital. RSV is highly infectious, and infection control measures, particularly good hand hygiene, are needed to prevent cross-infection to other infants in hospital.

Prognosis

Most infants recover from the acute infection within 2 weeks. However, as many as half will have recurrent episodes of cough and wheeze (see below). Rarely, usually following adenovirus infection, the illness may result in permanent damage to the airways (*bronchiolitis obliterans*).

Prevention

A monoclonal antibody to RSV (palivizumab, given monthly by intramuscular injection) reduces the number of hospital admissions in high-risk preterm infants. Its use is limited by cost and the need for multiple intramuscular injections.

Pneumonia

The incidence of pneumonia peaks in infancy and old age, but is relatively high in childhood. Pneumonia is a major cause of childhood mortality in resource-poor countries. It is caused by a variety of viruses and bacteria, although in over 50% of cases no causative pathogen is identified. Viruses are the most common cause in younger children, while bacteria are commoner in older children. In clinical practice it is difficult to distinguish between viral and bacterial pneumonia.

The pathogens causing pneumonia vary according to the child's age:

- Newborn – organisms from the mother's genital tract, particularly group B streptococcus, but also Gram-negative enterococci
- Infants and young children – respiratory viruses, particularly RSV, are most common, but bacterial infections include *Streptococcus pneumoniae* or *Haemophilus influenzae*. *Bordetella pertussis* and *Chlamydia trachomatis* can also cause pneumonia at this age. An infrequent but serious cause is *Staphylococcus aureus*
- Children over 5 years – *Mycoplasma pneumoniae, Streptococcus pneumoniae* and *Chlamydia pneumoniae* are the main causes.
- At all ages *Mycobacterium tuberculosis* should be considered.

A conjugate vaccine (Prevenar), with immunogenicity against thirteen of the most common serotypes of *Streptococcus pneumoniae* responsible for invasive disease, is now included in the routine immunisation schedule in the UK and many countries. There has been a marked reduction in the incidence of pneumonia from *Haemophilus influenzae* type B since the introduction of Hib immunisation.

Clinical features

Fever and difficulty in breathing are the commonest presenting symptoms, usually preceded by an upper respiratory tract infection. Other symptoms include cough, lethargy, poor feeding and an 'unwell' child. Localised chest, abdominal, or neck pain is a feature of pleural irritation and suggests bacterial infection.

Examination reveals tachypnoea, nasal flaring and chest indrawing – the best clinical sign of pneumonia in children is increased respiratory rate, and pneumonia can sometimes be missed if the respiratory rate is not measured in a febrile child (so-called 'silent pneumonia'). There may be end-inspiratory respiratory coarse crackles over the affected area, but the classic signs of consolidation with dullness on percussion, decreased breath sounds and bronchial breathing over the affected area are often absent in young children. Oxygen saturation readings may be decreased; this is an indication for hospital admission.

A chest X-ray may confirm the diagnosis, but with the exception of a classic lobar pneumonia

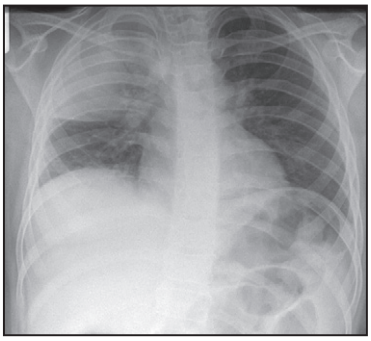


Figure 16.7 Consolidation of the right upper lobe. Lobar consolidation is a feature of pneumococcal pneumonia.

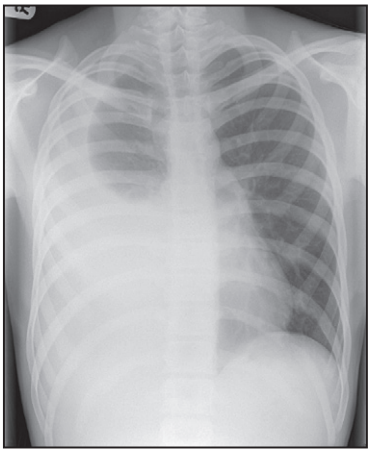


Figure 16.8 Right-sided empyema.

characteristic of *Streptococcus pneumoniae* (Fig. 16.7), a chest X-ray cannot differentiate between bacterial and viral pneumonia. In younger children, a nasopharyngeal aspirate is useful to identify viral causes, but blood tests, including full blood count and acute-phase reactants, are generally unhelpful in differentiating between a viral and bacterial cause. A small proportion of pneumonias are associated with a pleural effusion, where there may be blunting of the costophrenic angle on the chest X-ray. Some of these effusions develop into empyema and fibrin strands may form, leading to septations, which make drainage difficult (Fig. 16.8). The incidence of childhood empyema has risen over the last decade, the precise reason for which remains unclear. Ultrasound of the chest will often distinguish between parapneumonic effusion and empyema.

Management

Evidence-based guidelines for the management of pneumonia in childhood have been published (British Thoracic Society). Most cases can be managed at home, but indications for admission include oxygen saturation <93%, severe tachypnoea and difficulty breathing, grunting, apnoea, not feeding or family unable to provide appropriate care. General supportive care should include oxygen for hypoxia and analgesia if there is pain. Intravenous fluids should be given if necessary, to correct dehydration and maintain adequate hydration and salt balance. Physiotherapy has no role.

The choice of antibiotic is determined by the child's age, severity of illness and appearance on chest X-ray. Newborns require broad-spectrum intravenous antibiotics. Most older infants can be managed with oral amoxicillin, with broader-spectrum antibiotics such as co-amoxiclav being reserved for those who are complicated or unresponsive. For children >5 years of age, either amoxicillin or an oral macrolide such as erythromycin is the treatment of choice.

Parapneumonic effusions usually resolve with appropriate antibiotics, but the small proportion that develop an empyema require drainage of the collection. This may be achieved by either placement of a chest drain with or without the installation of a fibrinolytic agent in the intrapleural space (e.g. urokinase) to break down any septations, or by surgical decortication. Practices vary between different centres.

Prognosis

Follow-up is not generally required for children with simple consolidation on chest X-ray and who recover clinically. Those with evidence of lobar collapse, atelectasis or empyema should have a repeat chest X-ray after 4–6 weeks. Virtually all children with pneumonia, even those with empyema, make a full recovery.

Consider pneumonia in children with neck stiffness or acute abdominal pain.

Asthma

Asthma is the most common chronic respiratory disorder in childhood, affecting 15–20% of children. Worldwide there has been a significant increase in the incidence of asthma over the last 40 years, although this has now plateaued in many developed countries. Although the symptoms of asthma are readily controlled in most children, it is an important cause of absence from school, restricted activity and anxiety for the child and family. There are still about 20 deaths from asthma in children each year in the UK.

Diagnosing asthma in preschool children is often difficult. Approximately half of all children wheeze at some time during the first 3 years of life. In general, there are two patterns of wheezing (Fig. 16.9):

- Transient early wheezing
- Persistent and recurrent wheezing.

Transient early wheezing

Most wheezy preschool children have *virus-associated wheeze* (also known as *episodic viral wheeze* and *wheezy bronchitis*). Transient early wheezing is thought to result from small airways being more likely to narrow and obstruct due to inflammation and aberrant immune responses to viral infection. This gives the